

3D Volume Image Analysis for Biologists

Introduction to image formation and basics of Fiji / Napari

NUS National University of Singapore
Microscopy Cluster
Yong Loo Lin School of Medicine



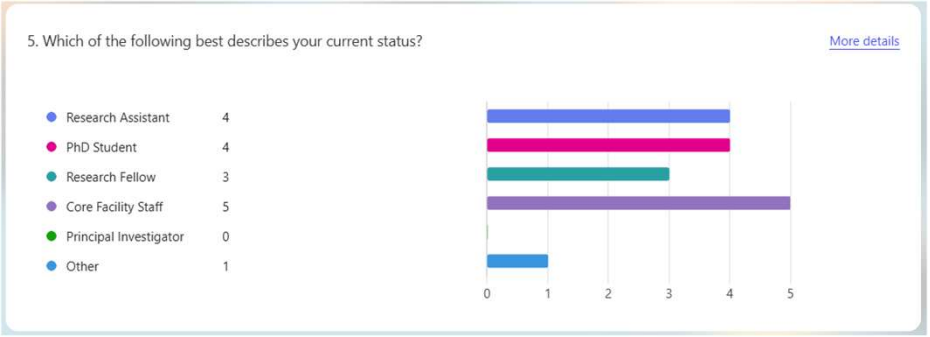
Samaksh Singh
Computer Scientist
3+ years of experience in HPC and EM analysis

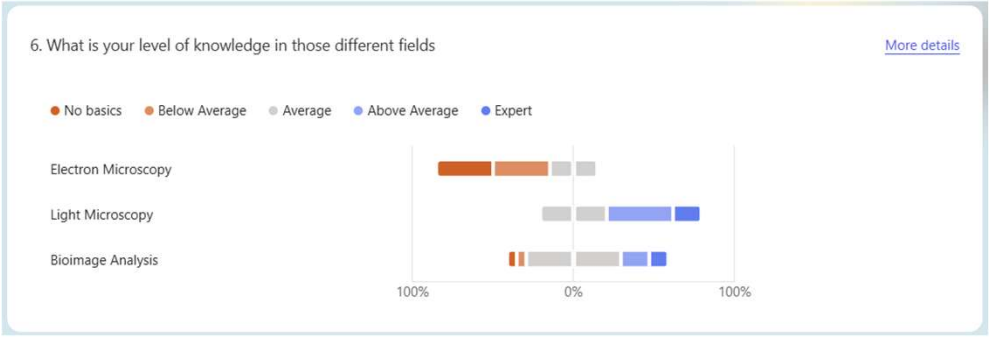


Lee Shu Ying
Senior Lab manager
17+ years of experience in LM



Richard De Mets
Deputy Director
10+ years of being a Swiss knife





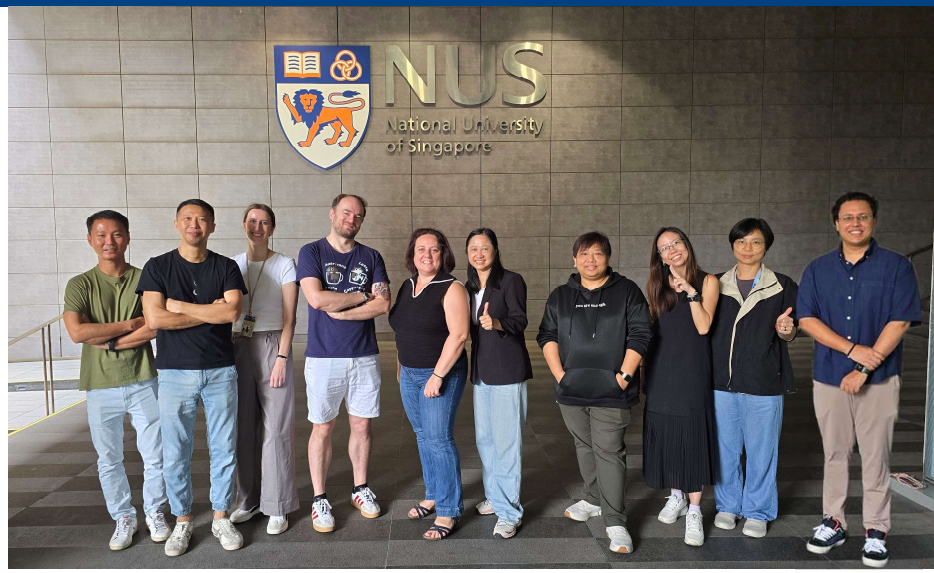
NUS National University of Singapore | Microscopy Cluster
Yong Loo Lin School of Medicine



Content

- Basics of Fiji
- Anaconda essentials
- Basics of Napari
- LLMs for Bioimage Analysis

General introduction

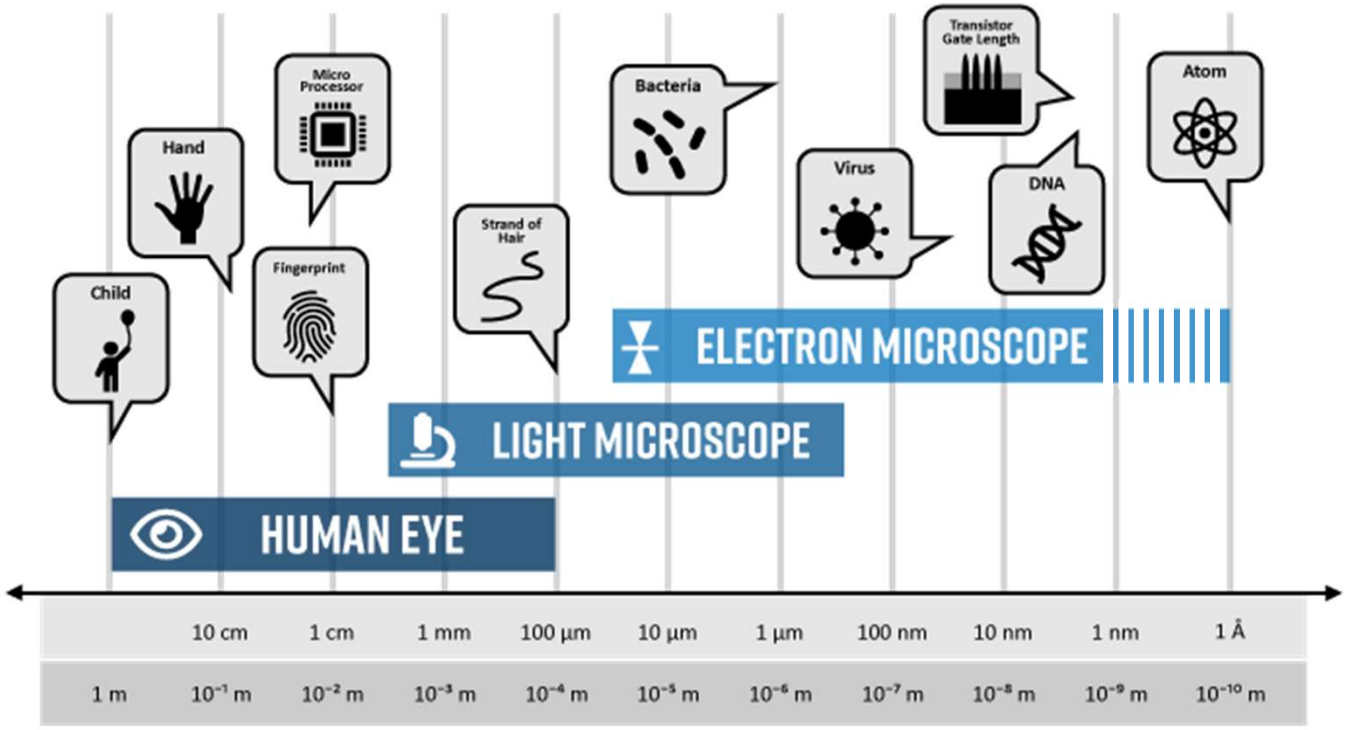


We provide expert guidance and high quality state-of-the-art imaging solutions for biomedical researchers in the Yong Loo Lin School of Medicine and the scientific community.

We partner with researchers throughout their entire scientific journey, championing best practices in image acquisition and analysis to enable robust and reproducible results.

General introduction

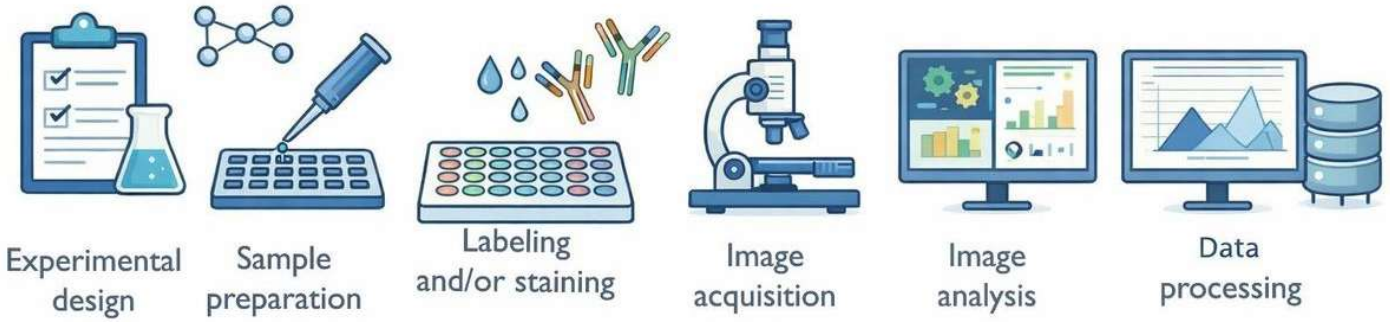
We provide expert guidance and high quality state-of-the-art imaging solutions for biomedical researchers in the Yong Loo Lin School of Medicine and the scientific community.



General introduction

We partner with researchers throughout their entire scientific journey, championing best practices in image acquisition and analysis to enable robust and reproducible results.

Accompanying researchers at each step of their research project



Light Microscopy Unit



Lee Shu Ying
Senior Lab manager



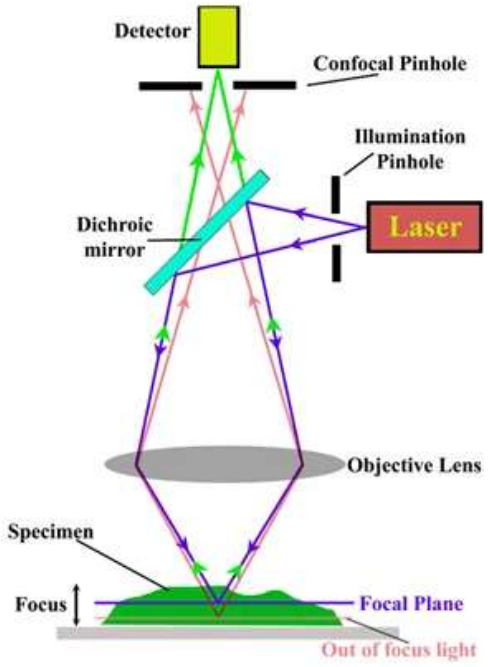
Delia Pang
Senior Lab executive

	Name of Equipment	Year Installed	Remarks
1	Yokogawa CQ1	2024	Confocal Spinning-disk, High-throughput imaging and analysis
2	Olympus FV3000 (Resonant)	2019	Workhorse Confocal with resonant scanning & live imaging
3	Olympus FV3000	2021	Workhorse Confocal
4	Zeiss Axioscan 7	2026	Automated 100 slide loader, with color camera for H&E and fluorescence for IHC
5	EVOS M7000	2022	Routine slide and plate imaging
6	TissueFAXS (repurposed from old LSM510)	2016	4-slide slide scanner, with color camera for H&E and fluorescence for IHC
7	Zeiss AxioObserver	2009	Dedicated for live cell imaging
8	Leica SP8 Multiphoton	2020	Multiphoton Microscope for deep imaging, label-free and Fluorescence Lifetime

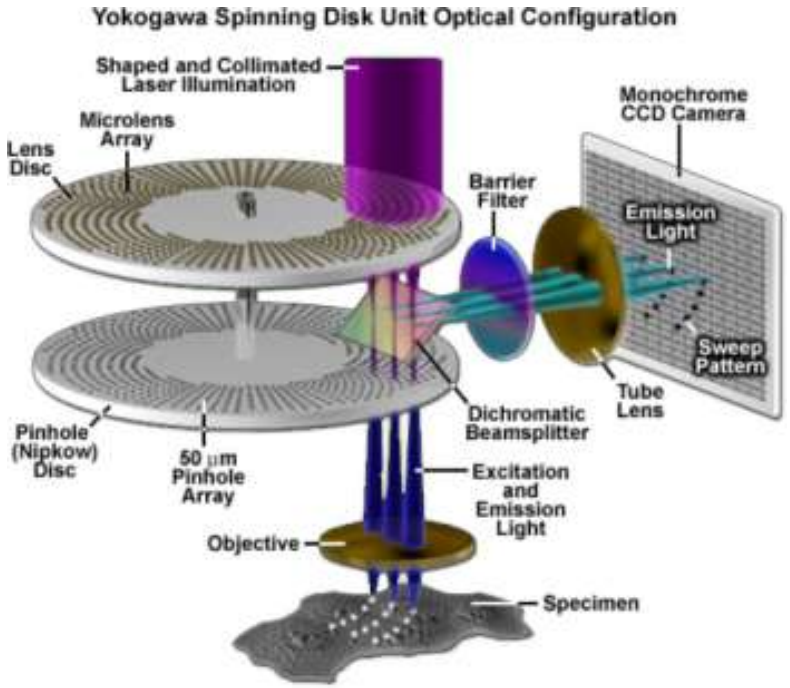
Confocal Microscope
Brightfield Microscope
Multiphoton Microscope

Light Microscopy Unit

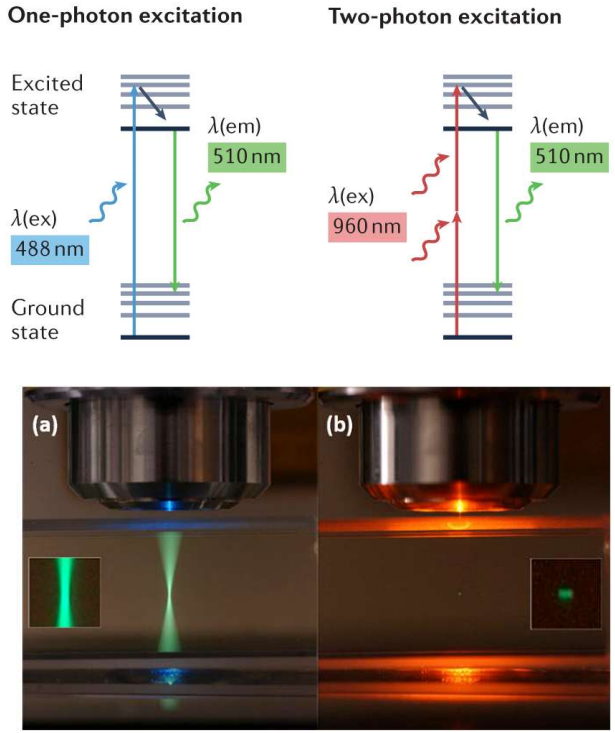
Confocal principles



Laser Scanning Confocal



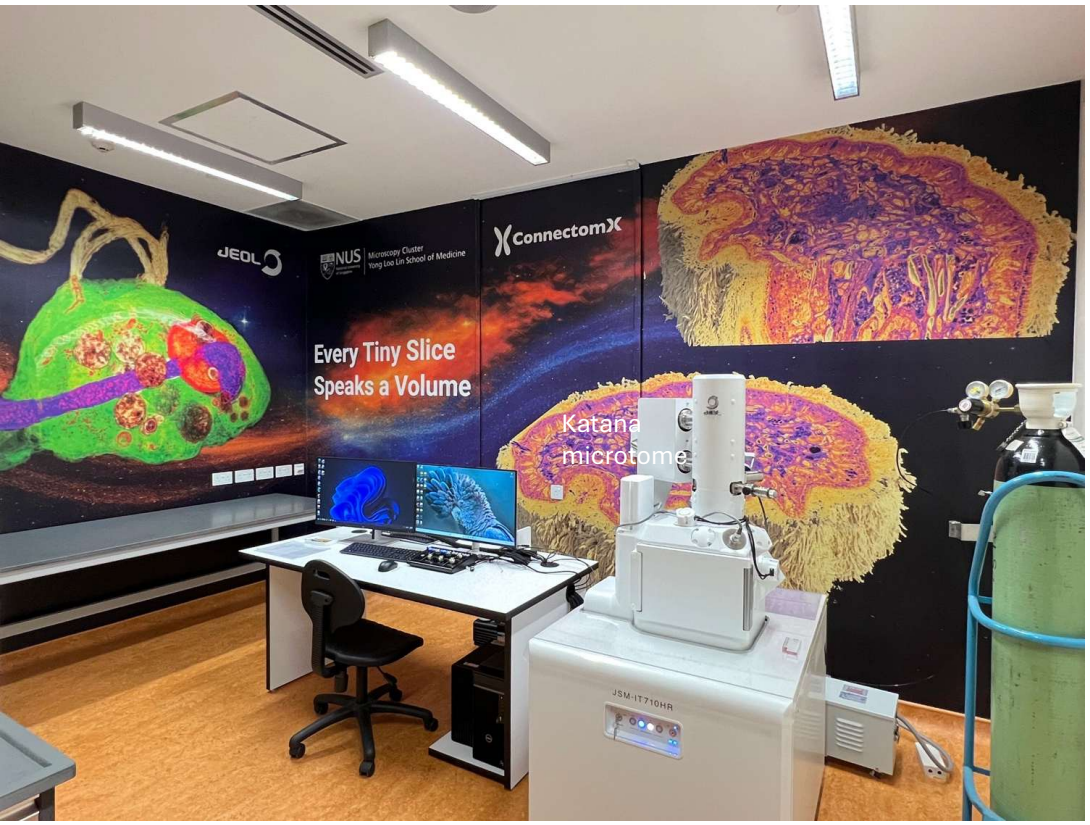
Spinning Disk Confocal



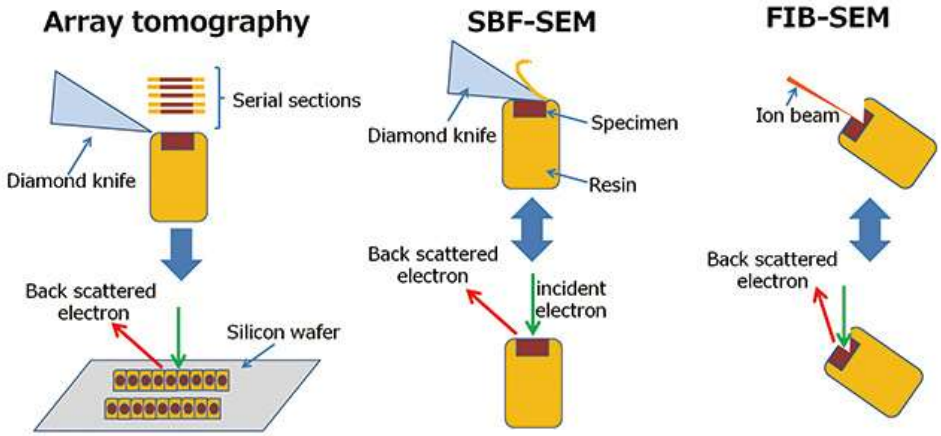
MultiPhoton LSM

Electron Microscopy Unit

- volume EM at NUS YLL Microscopy Cluster



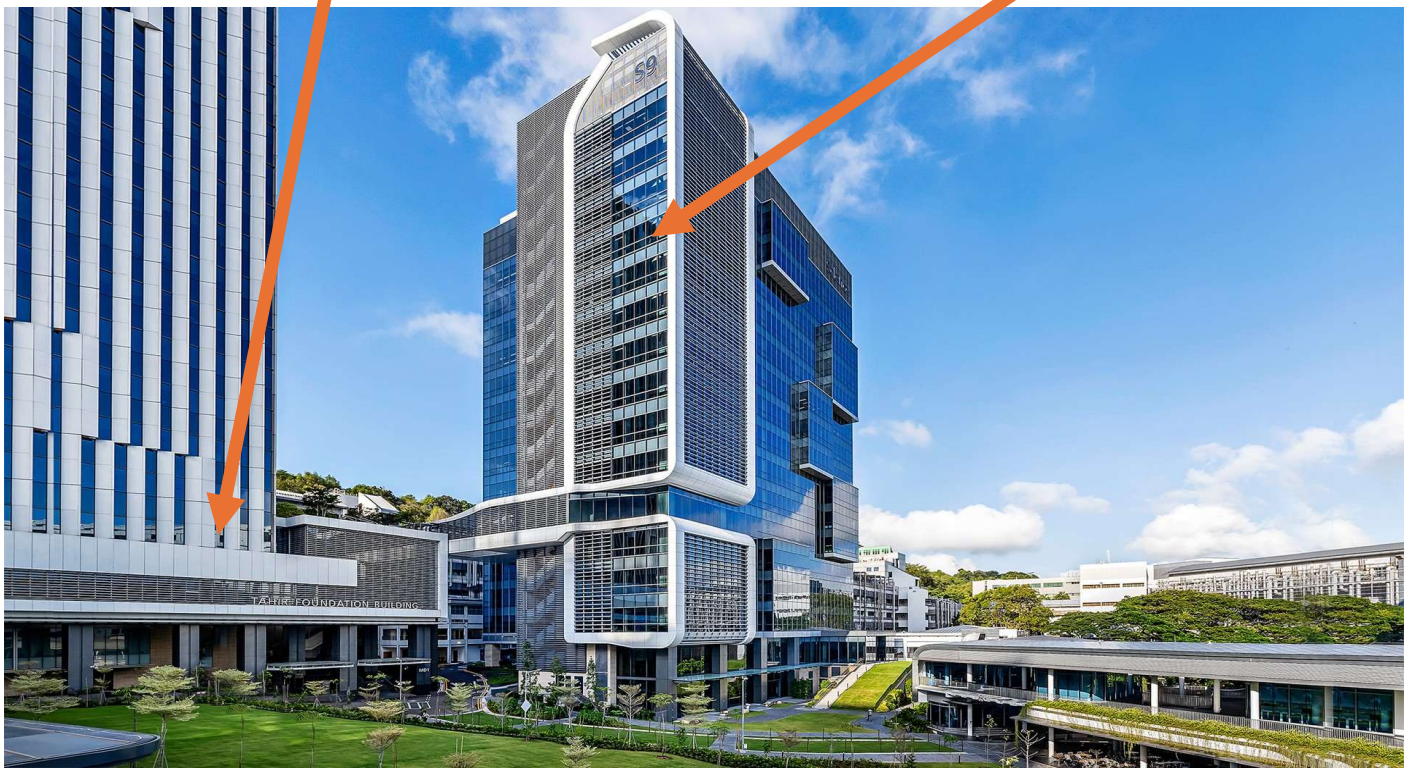




General introduction

LMU and EMU @ MD1 #B1

Multiphoton and Histology @S9 level 11



Key Contact(s)

LMU:

Lee Shu Ying
medleesy@nus.edu.sg

Delia Pang
deliapang@nus.edu.sg

EMU:

Micky Leong
medlln@nus.edu.sg

Isabelle Bonne
i.bonne@nus.edu.sg

Free software

CELLPOSE
CELLPROFILER
FIJI
BLENDER
iMOD
IMOD
ICY
ILASTIK
NAPARI
QUPATH
ZEROCOSTDL4MIC
R STUDIO

Commercial software

AMIRA-AZIVO
ARIVIS
IMARIS
VISIOPHARM
DRAGONFLY
CELLSENS
MATLAB
HUYGENS
ZEN
AIVIA
METAMORPH

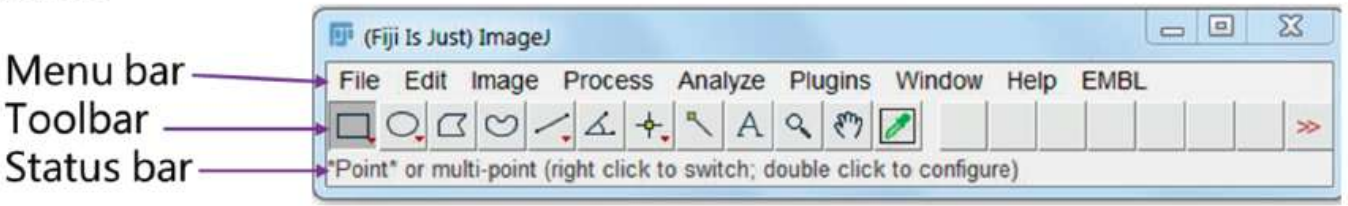
A Hitchhiker's guide through the bio-image analysis software universe, Haase & al, FEBS Letter 2022

Basics of Fiji

START FIJI

by clicking on the file:  which is in the downloaded folder.

Fiji opens:

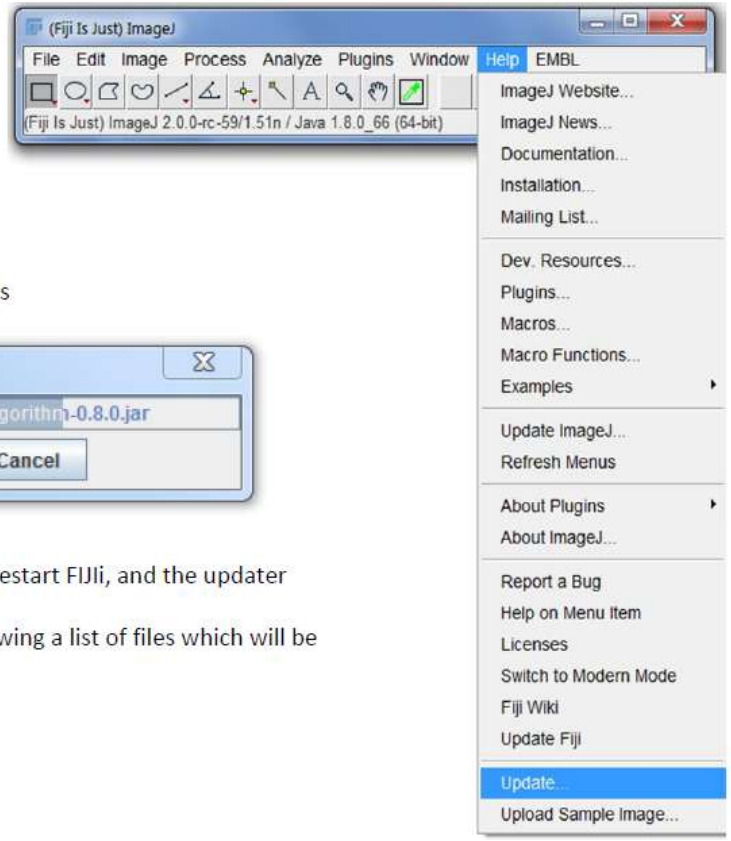


Plugin installation

UPDATE...

Go to: Help > Update...

!!!! not Update fiji !!!!



A window will open checking for updates

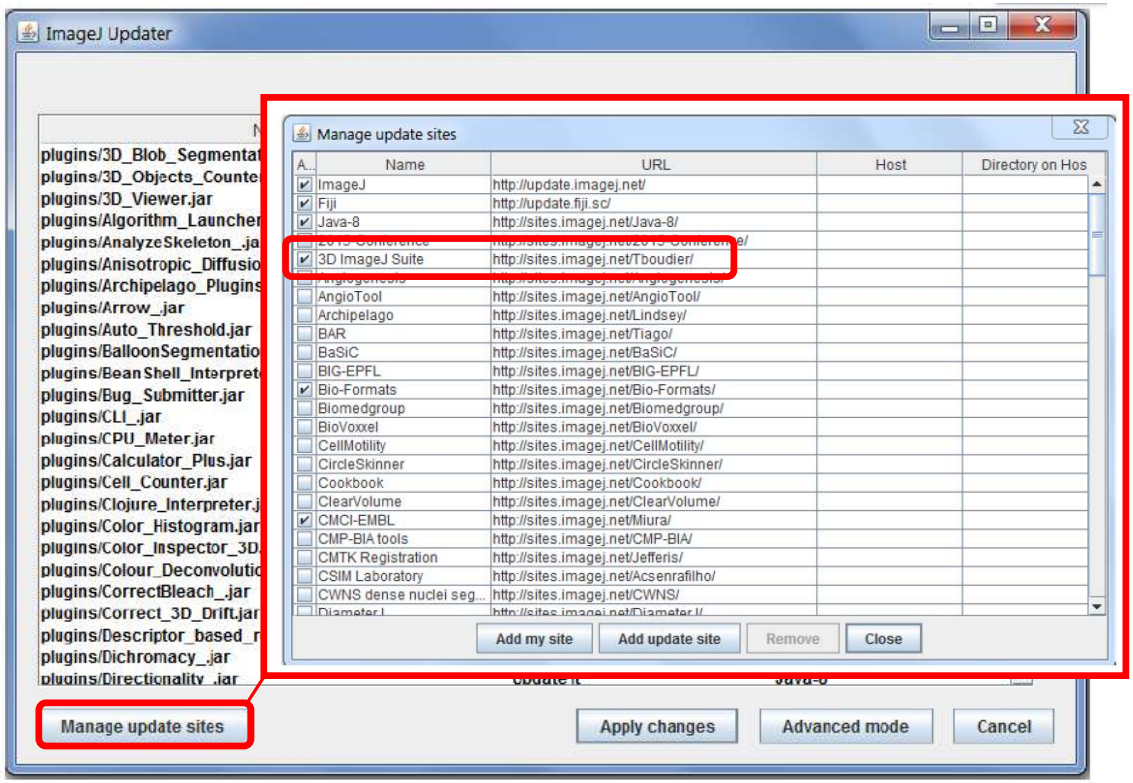


Follow the prompts if you are asked to restart FIJIi, and the updater

The ImageJ Updater will then open, showing a list of files which will be updated.

Plugin installation

1. Fiji-Cellpose
2. SAMJ
3. ImageScience
4. Tensorflow
5. Stardist
6. PTBIOP
7. IJPB-plugins

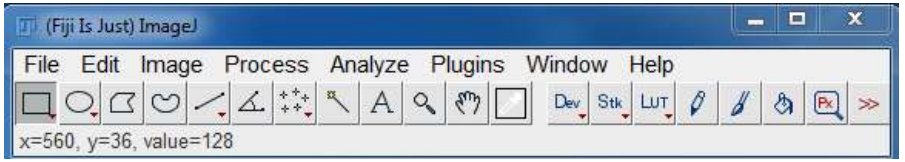


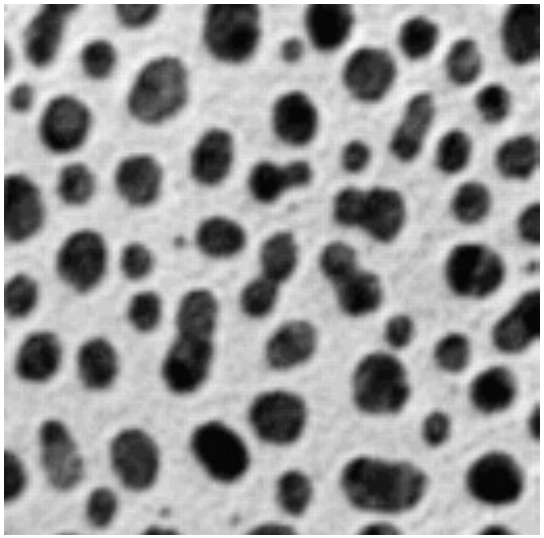
Click on “manage update sites” at the bottom right of the ImageJ updater.

RESTART ImageJ NOW

An image for a computer

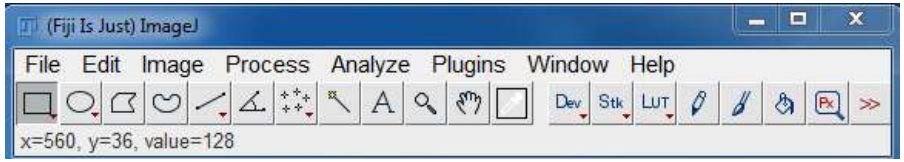
1. File > Open Samples > Blobs
2. Navigate the image with the mouse
3. Read the values in fiji bar.



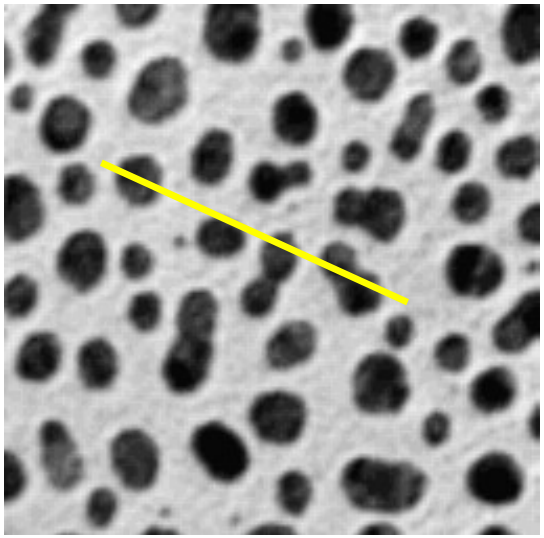


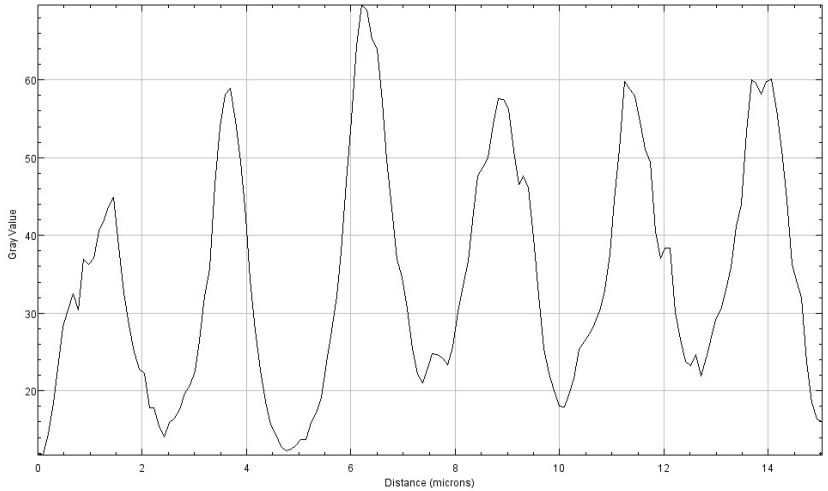
35	41	47	44	54	58	54	52	41	41
30	34	36	42	50	50	49	48	43	40
30	30	41	48	61	65	73	68	60	53
38	44	54	60	72	88	99	98	86	73
47	57	66	67	81	94	108	110	100	80
57	66	68	78	76	87	84	90	79	79
49	58	62	64	64	62	57	62	59	59
36	38	43	46	53	58	55	60	59	51
23	29	35	43	50	63	73	74	72	57
25	27	37	43	57	63	67	76	71	63

Intensity Profile



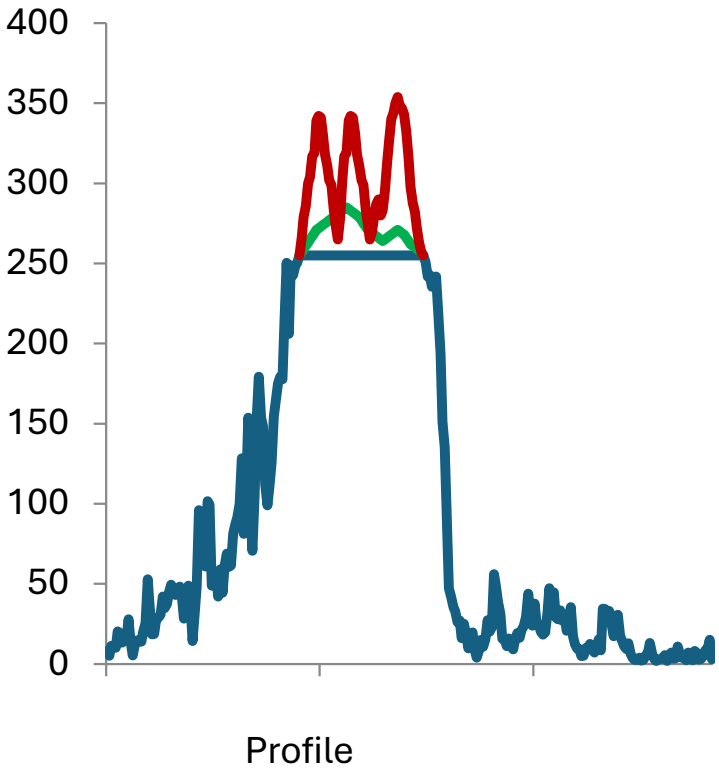
- Draw a line with the line tool across a structure
- Plot the intensity profile: Analyse > Plot profile (Ctrl + K)

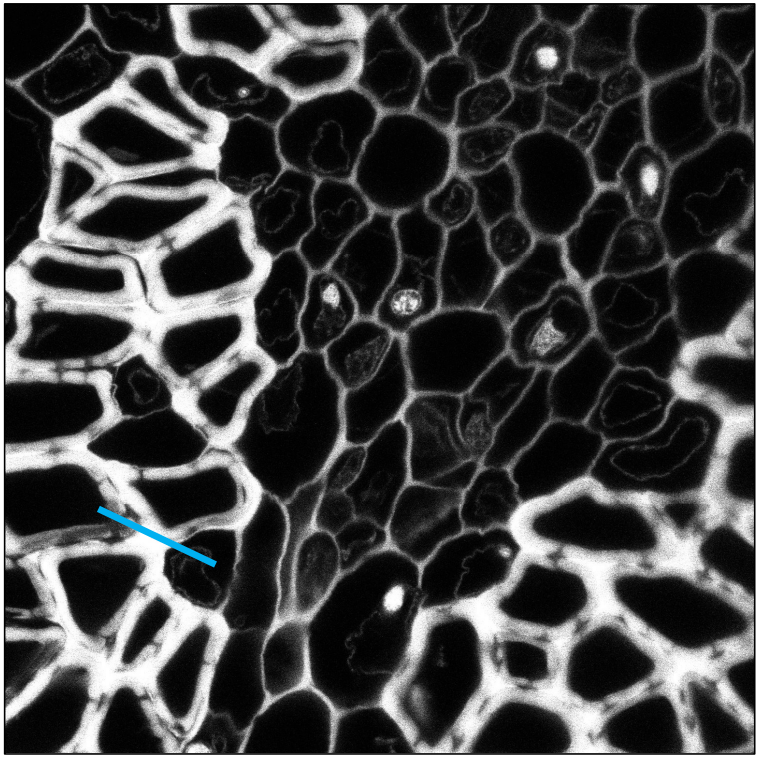




LUT is inverted !!

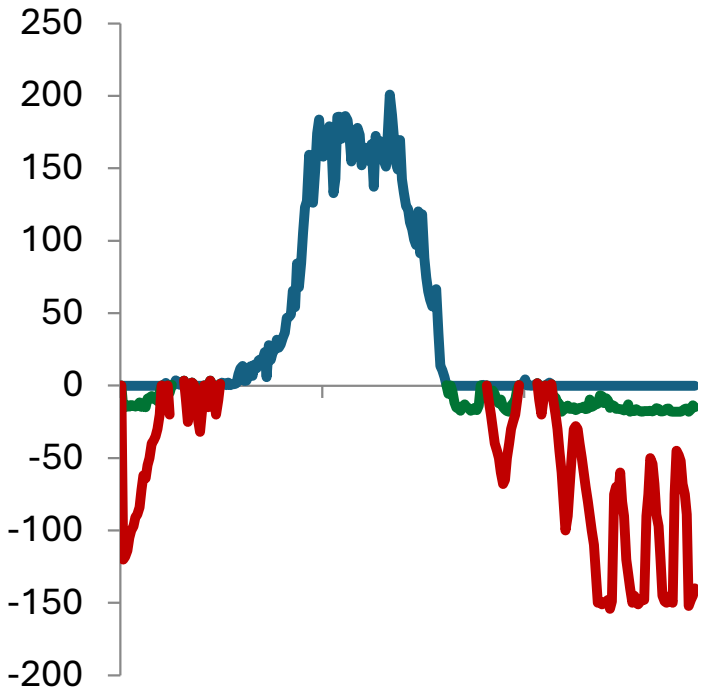
Intensity distribution - Saturation



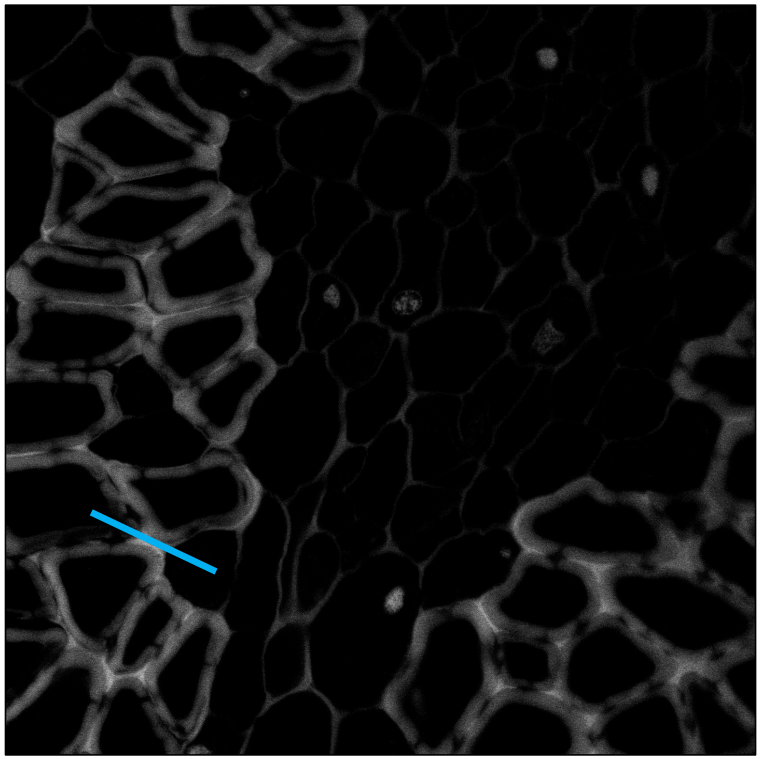


Adapted from P. Bankhead

Intensity distribution – Clipped (underexposed)



Profile

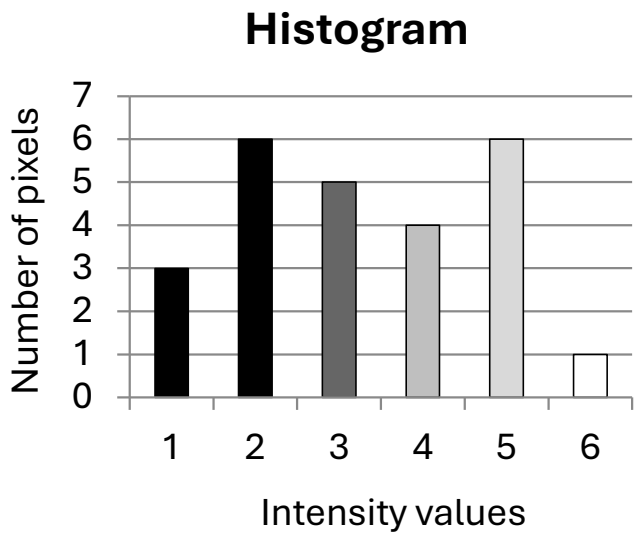


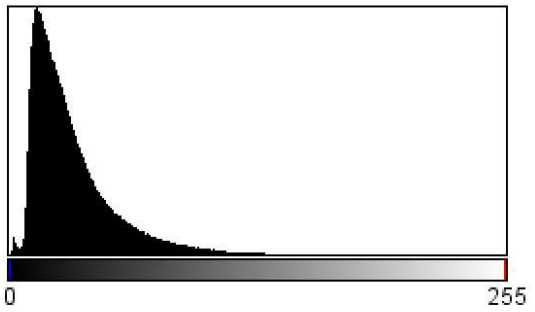
Adapted from P. Bankhead

Image Histogram

Frequency of occurence of a given pixel value

1	2	3	4	5
3	1	5	3	2
2	5	6	1	5
5	3	4	2	3
4	2	2	5	4





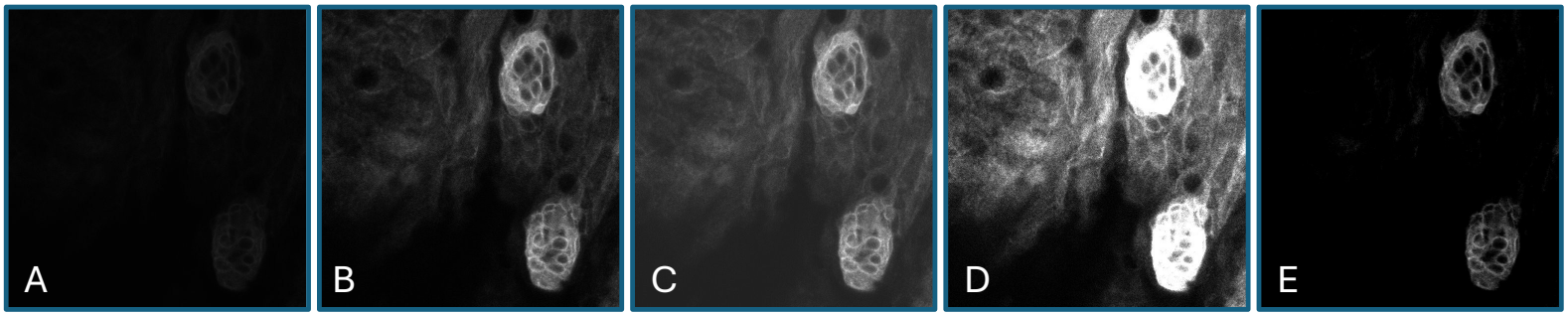
Count: 609961
Mean: 32.686
StdDev: 23.391
Min: 1
Max: 252
Mode: 14 (20325)

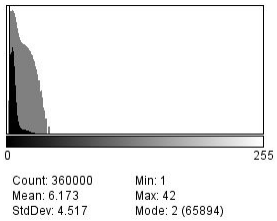
Image Histogram

- Open the files A to E from the folder ”Dynamic Range”
- Examine the pixel values by all the techniques you know

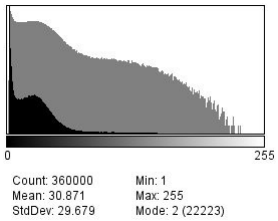
- Which of the images is best described by
  - Low intensity values not used ?
  - High intensity values not used?
  - Clipped?
  - Saturated?
  - Good use of the dynamic range?

Image Histogram

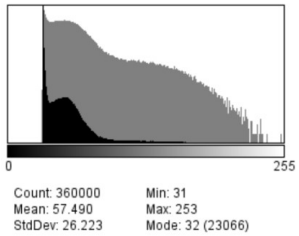




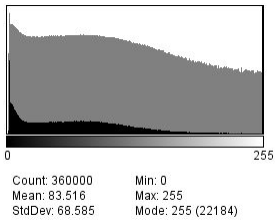
High intensity values not used



Good use of dynamic range

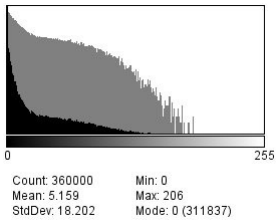


Low intensity values not used



Saturated

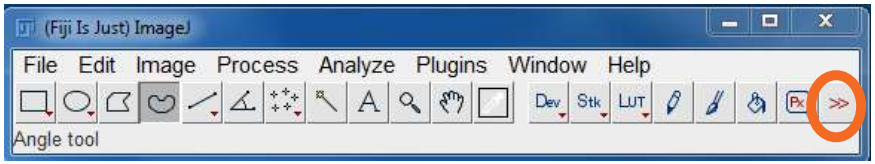




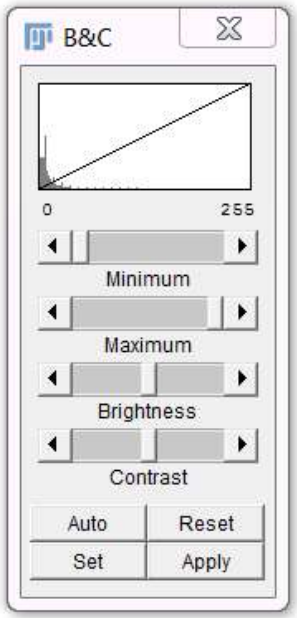
Clipped



Image Display

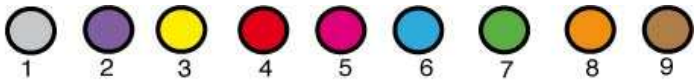


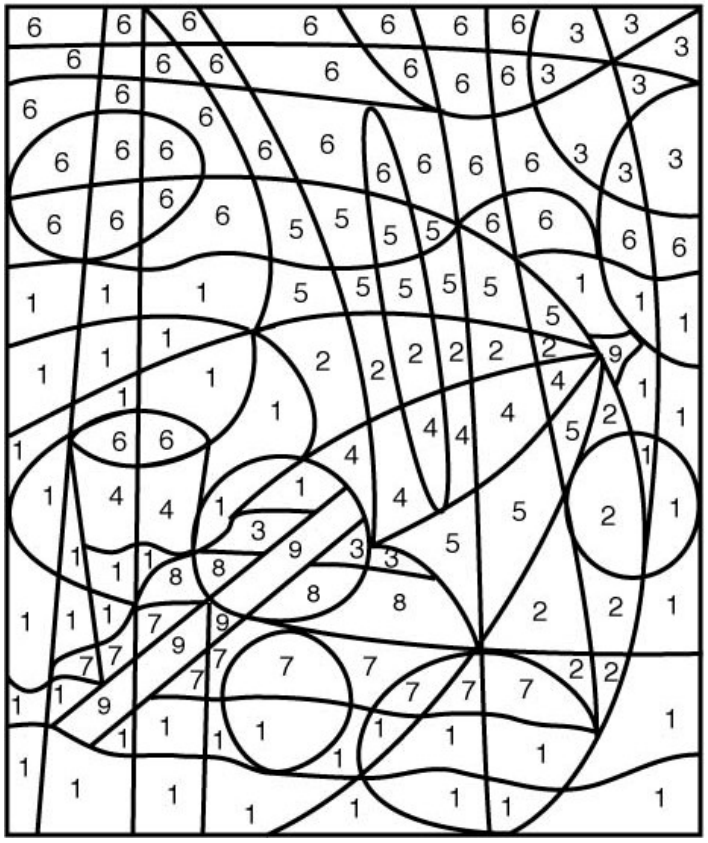
- Open the File > Image Samples “Blobs” (Invert LUT for ease of use)
- Select the *Pixel inspector* to view intensity values in a small region of the image (Find it under >>), and click somewhere on the image
- *Open the B&C manager*
  Image > Adjust > Brightness/Contrast…
- Adjust the minimum and maximum sliders
- Check pixel values, do you see any changes?
- This time press ”*Apply*” in B&C manager
- Check pixel values, do you see any changes?

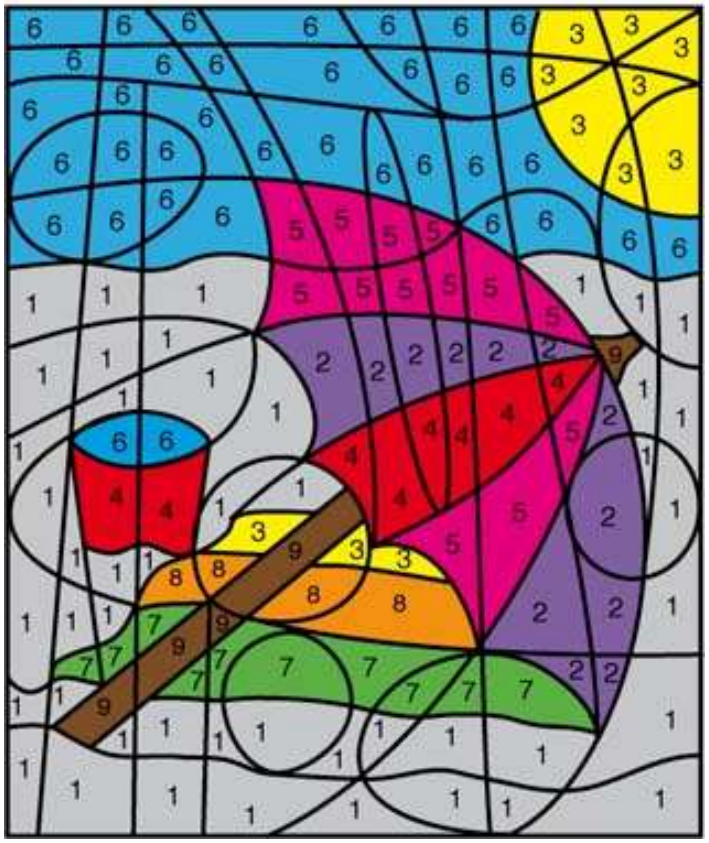


! Use the *Image > Duplicate* tool to keep the raw data safe

Lookup Tables

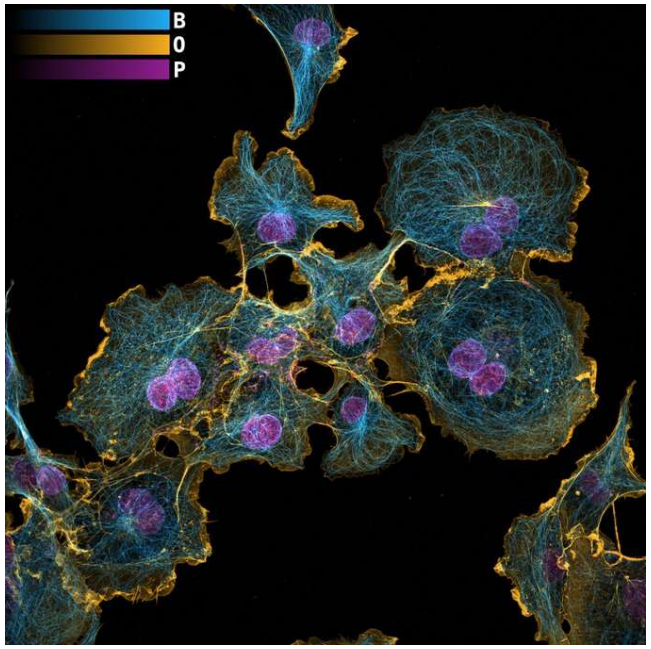


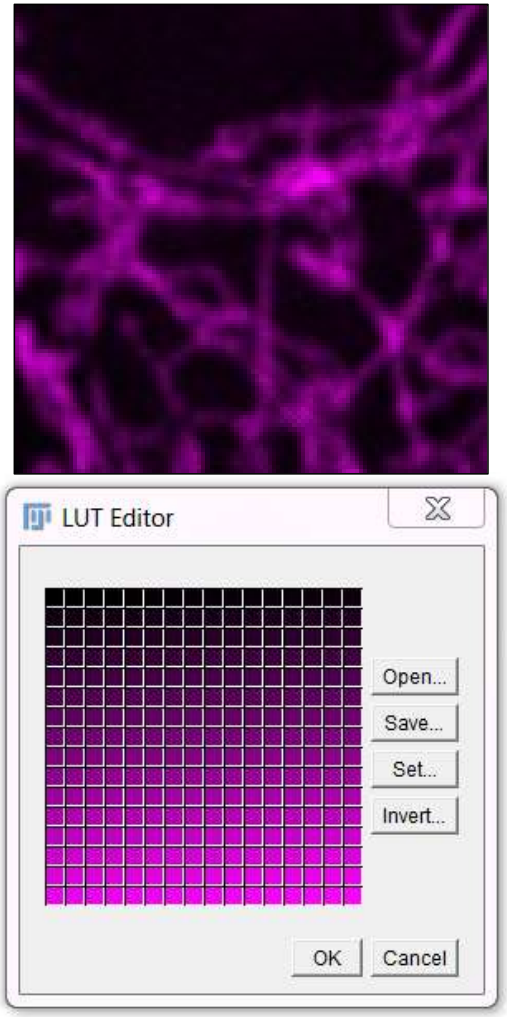




Lookup Tables

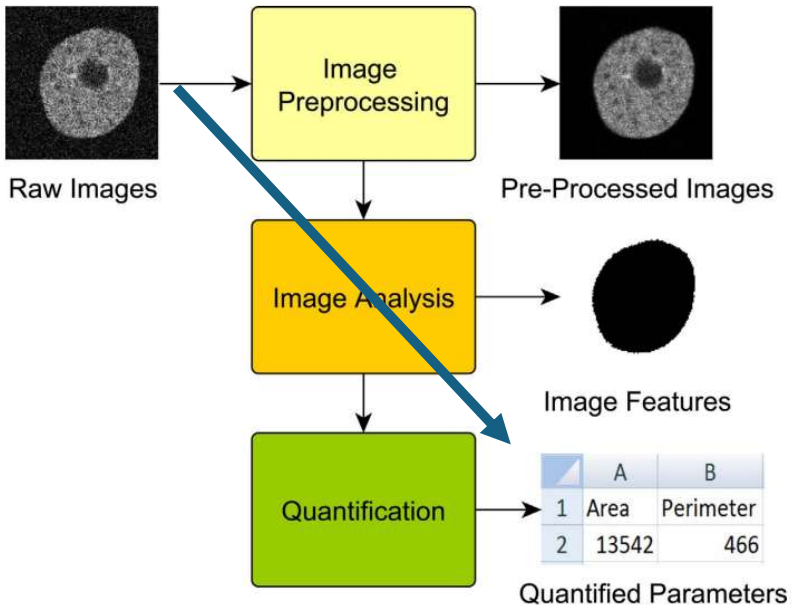
- Display and Edit
  Image > Color > Edit LUT
- Display the definition
  Image > Color > Show LUT





Typical Image Analysis workflow :

- Preprocessing – Denoising, normalization
- Segmentation – Thresholding, Machine/Deep Learning
- Measurement

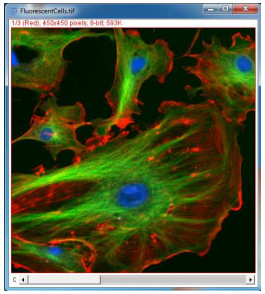


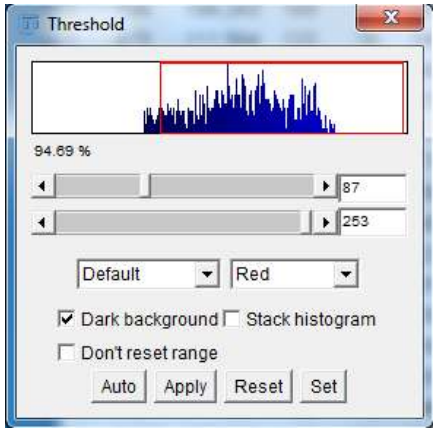
- Restore: denoising, background removal, deconvolution
- Detect/Segment: binarization, labels, object classification
- Measure: shape, intensities, distances

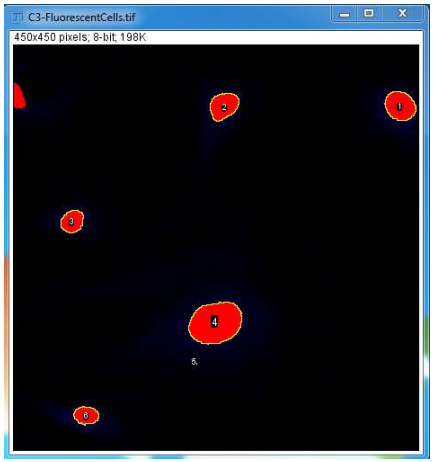
Wollmann & al, Journal of Biotechnology (2017) doi.org/10.1016/j.jbiotec.2017.07.019

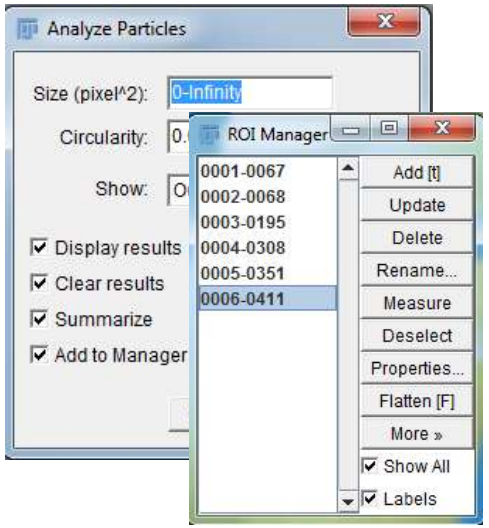
Thresholding and Analyzing objects

- Open the file “FluorescentCells”
- Go to Image → Color → Split channels
- Select the DAPI (nuclei in blue) channel
- Duplicate the image using Image → Duplicate
- Go to Process → Filters → Gaussian Blur (sigma of 2, press ok)
- Go to Image → Adjust → Threshold
- Set maximum and minimum intensity values (keep the window open, don’t click Apply)
- Open the ROI manager (Analyze → Tools → ROI Manager)
- Go to Analyze → Set measurements
- Tick Area and Mean Gray Value
- Redirect to the untouched DAPI image
- Go to Analyze → Analyze particles



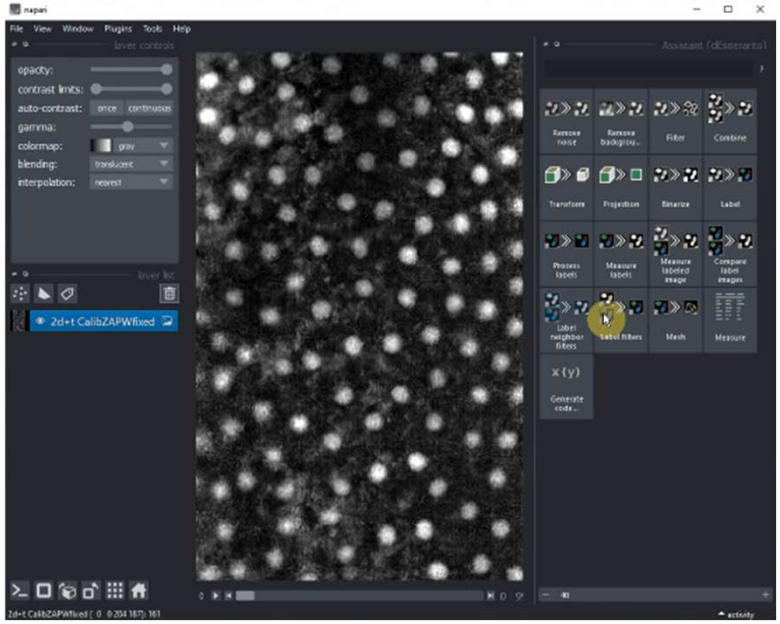


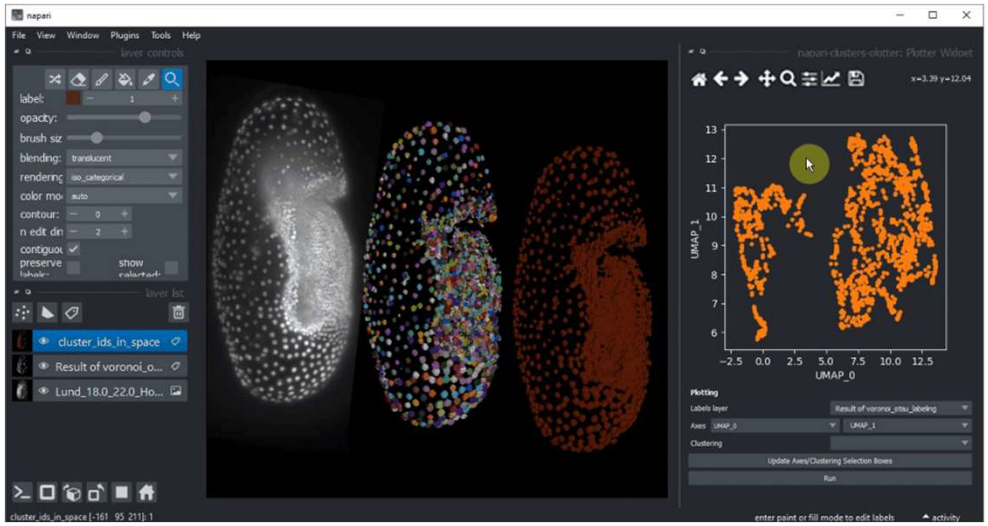




Questions ?

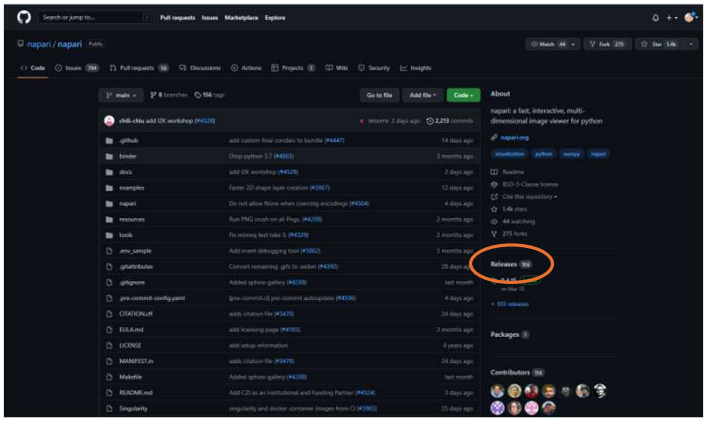
Napari – n-dimensional viewer in Python

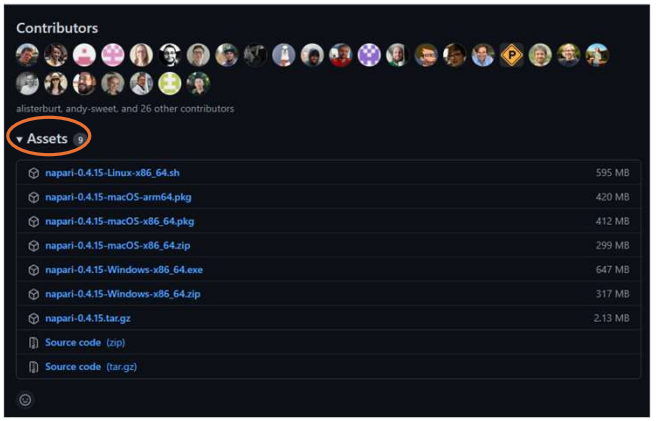




Napari – n-dimensional viewer in Python

- For regular usage:





https://github.com/napari/napari

- For developers (Python):

ANACONDA.

python™ Package Index

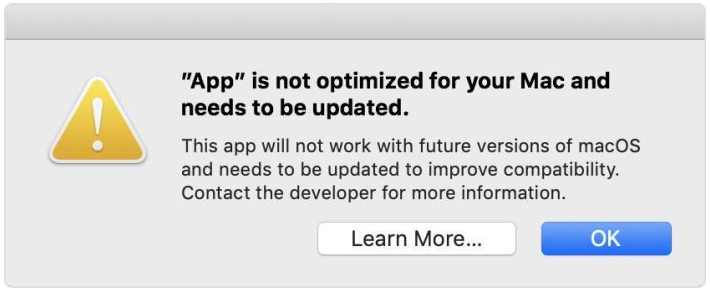
CONDA-FORGE

https://pypi.org/project/napari/

https://anaconda.org/conda-forge/napari

Anaconda – Python environment manager

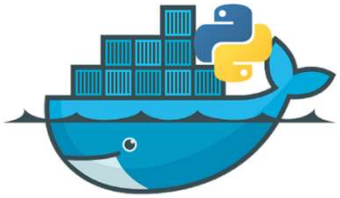


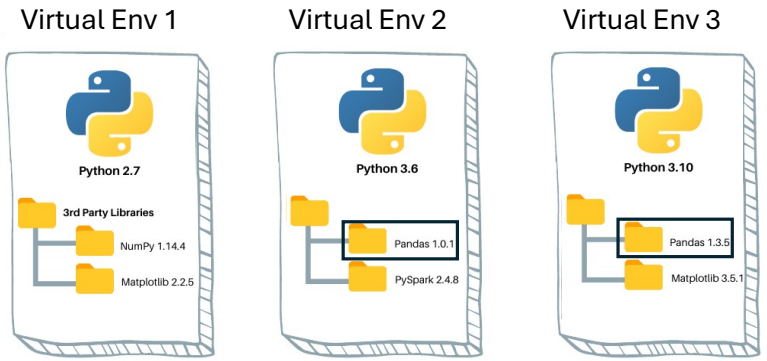


ANACONDA®

Python Virtualenv

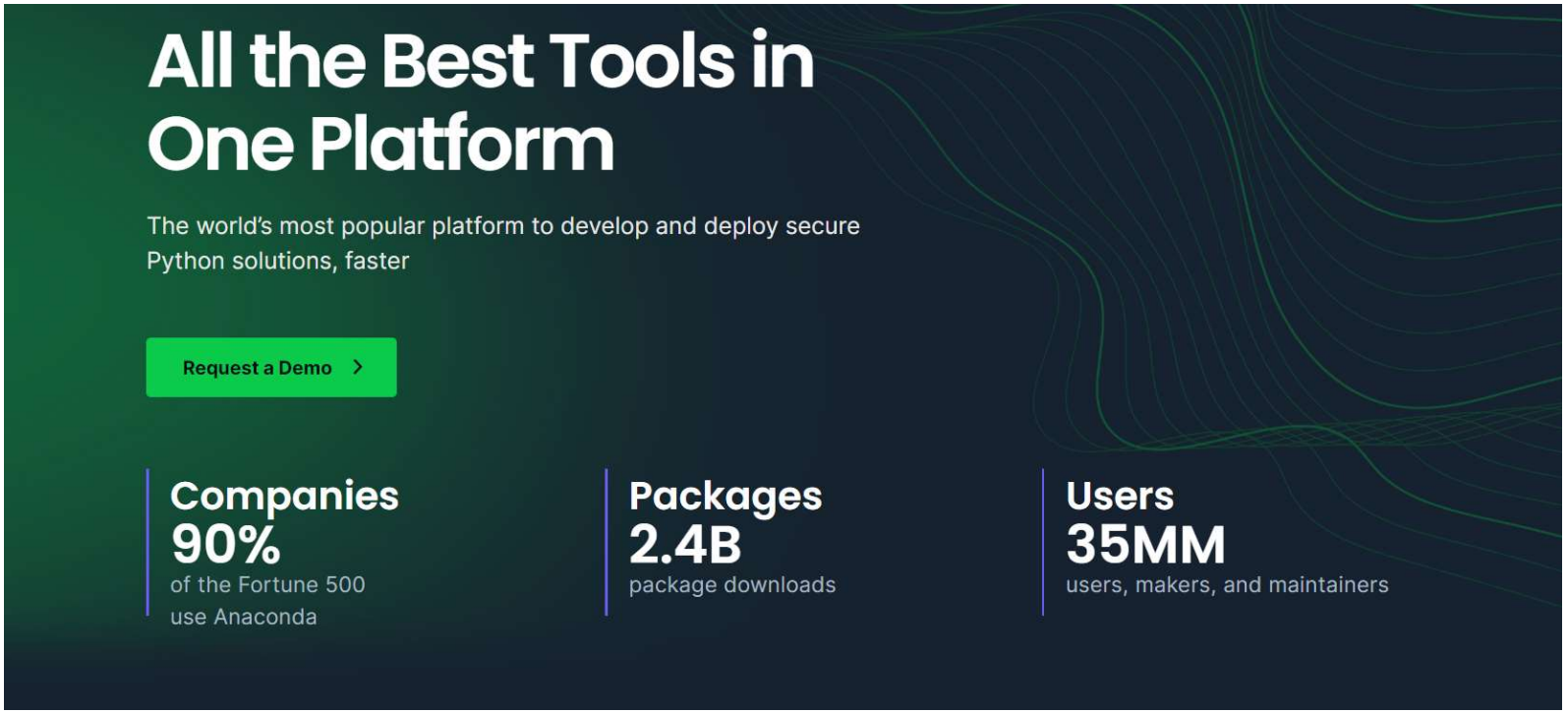
Poetry





dataquest.io

Why Anaconda

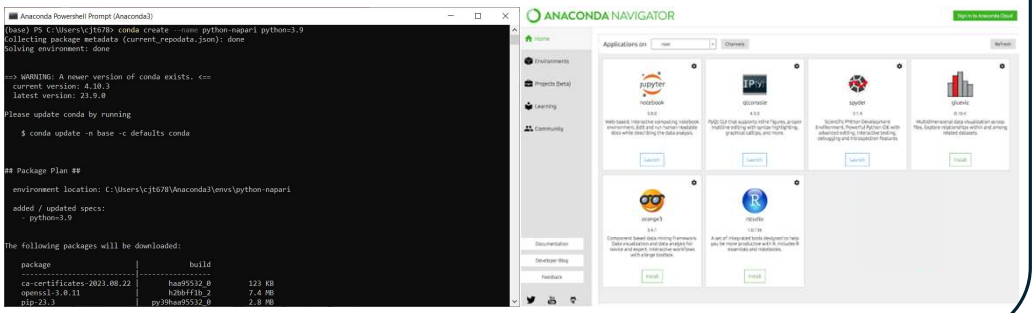


Why Anaconda

ANACONDA®

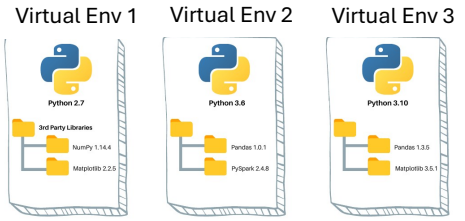
Anaconda supports both Python and R programming languages for scientific computing, that aims to simplify package management and deployment.

Virtual environment manageable with prompt or GUI



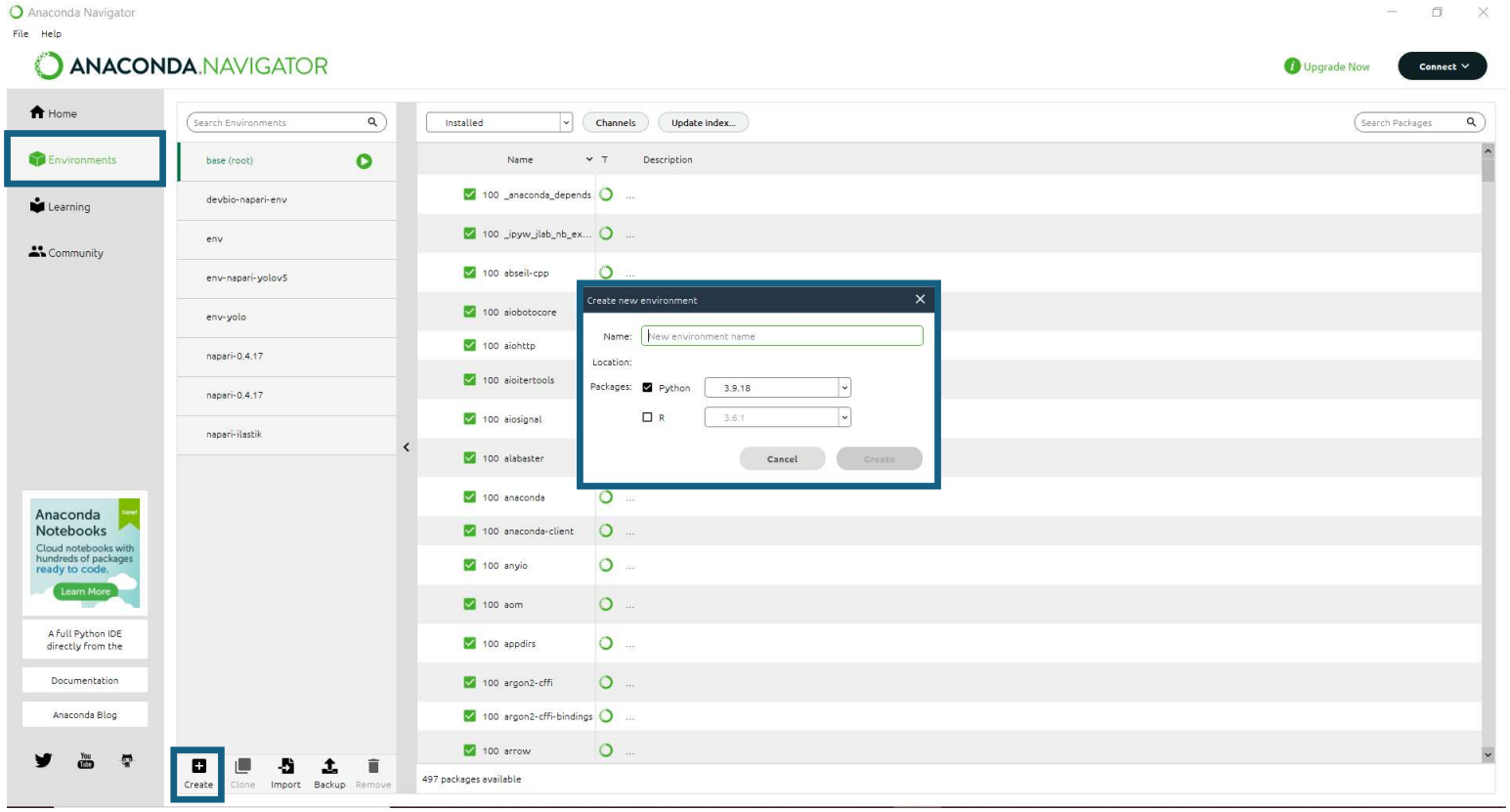
Repo of recipes packages in multiple languages

CONDA-FORGE



dataquest.io

Anaconda Navigator

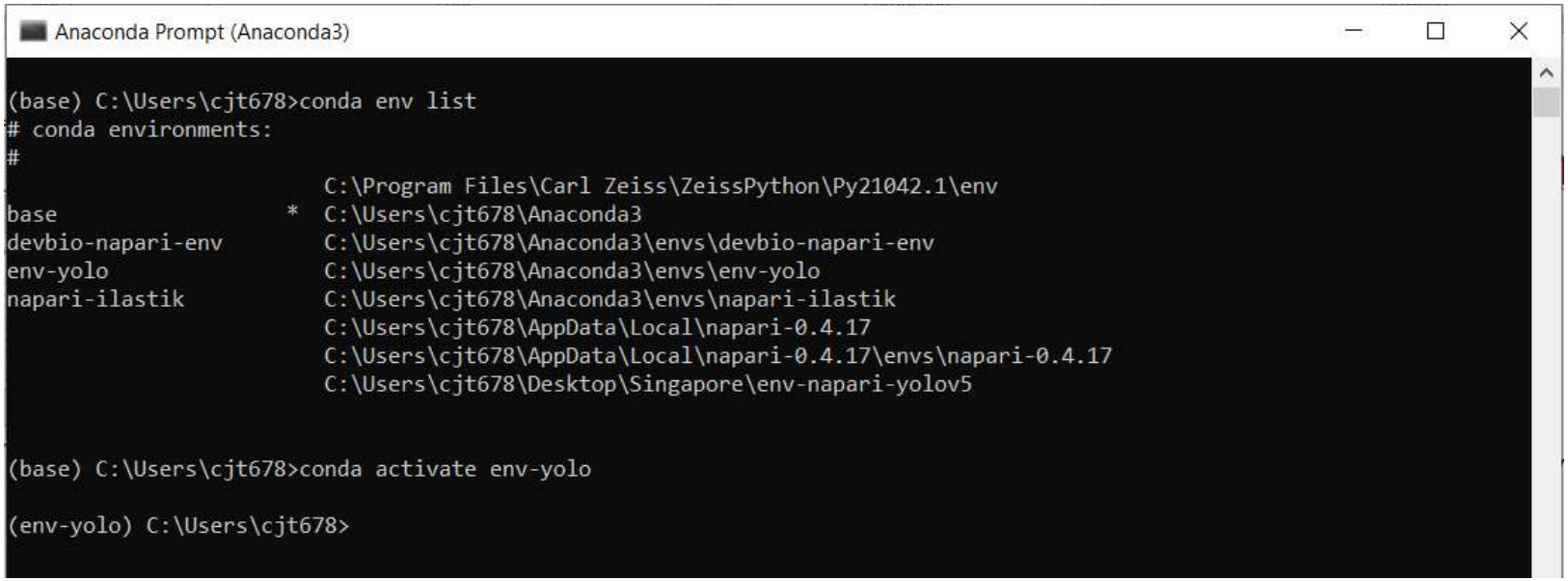


Anaconda Prompt - Create

- Classical
  - Conda create --name <environmentName> python=3.12
    - Example: conda create --name env-napari python=3.12
- Specify location
  - Conda create --prefix <path> python=3.12
    - Example: conda create --prefix C:\Users\Public\Desktop\conda\env-napari python=3.12
- From yaml file
  - Conda create env --file <file.yaml>
    - Example: conda create env --file C:\Users\Public\Desktop\devbio-napari.yaml

```yaml
name: sampleenv
channels:
  - conda-forge
dependencies:
  - python=3.9
  - scikit-learn=0.20.3
  - pillow=5.4.1
  - flask
  - pip:
    - numpy=1.16.2
    - pandas=0.24.*
    - plotly=3.7.0
```

Anaconda Prompt - Activate



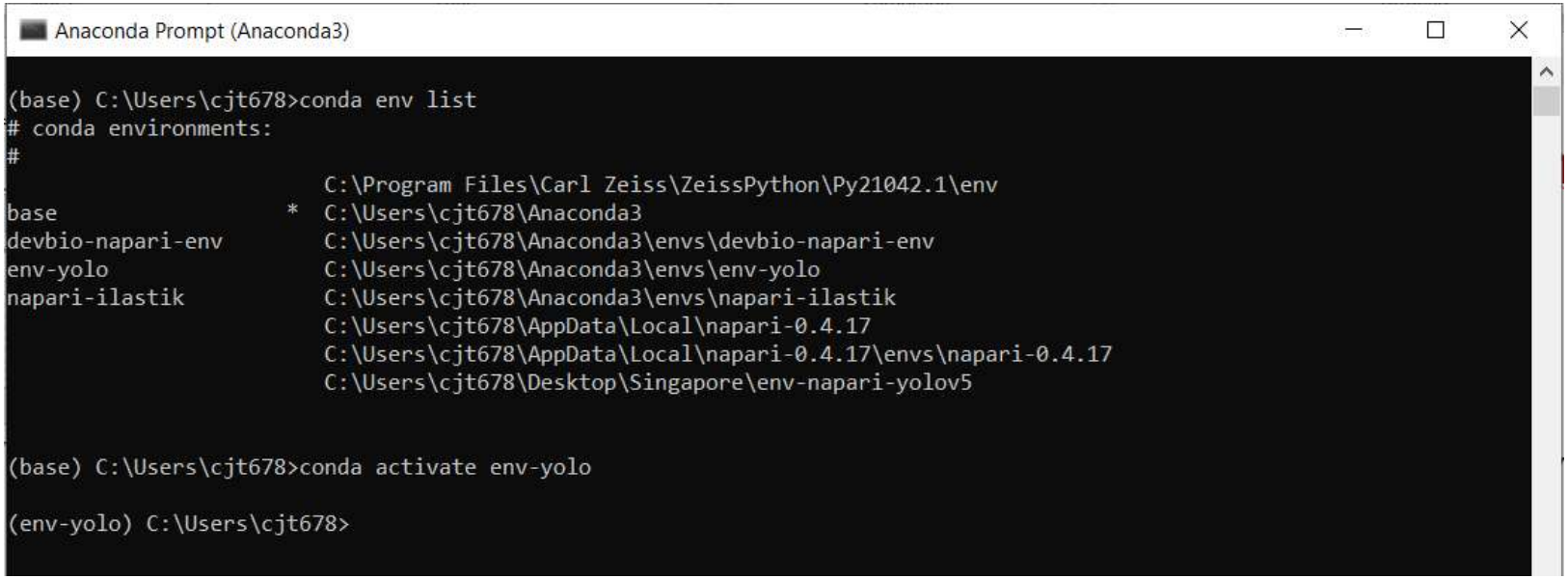
Conda activate <environment_name>

Manual installation : <package manager> install <package name>
Automated installation : <package manager> install --requirement requirements.txt

Anaconda Prompt - Install

```
napari-plugin-engine>=0.1.4
npe2 to <0.7.0 # https://github.com/haesleinhuepf/devbio-napari/issues/41
numpy>=1.21.4
napari-pyclesperanto-assistant
napari-skimage-regionprops
napari-animation
PlatyMatch
napari-plot-profile
napari-accelerated-pixel-and-object-classification
napari-brightness-contrast
napari-plugin-search
napari-segment-blobs-and-things-with-membranes
napari-simpleitk-image-processing
napari-folder-browser
napari-crop
napari-clusters-plotter>=0.7.1
napari-stl-exporter
napari-tabu
napari-workflow-optimizer
napari-workflow-inspector
napari-curtain
napari-layer-details-display
# https://github.com/napari/napari/issues/4415
# https://github.com/napari/napari/issues/4708
napari
vispy
napari-mouse-controls
the-segmentation-game
napari-process-points-and-surfaces
napari-blob-detection
jupyterlab
napari-czifile2
napari-roi
pydantic!=1.10.0 # https://twitter.com/Czaki_PL/status/1564908046186434566?s=20&t=YEAgvsUTuYIDRYlKREQp3A
```

Anaconda Prompt - Install



(Make sure that you are not in the **base** environment anymore)

```
conda activate env-napari
pip install --requirement https://raw.githubusercontent.com/rdemets/volume-course-2026/refs/heads/main/requirements.txt
```

Anaconda Prompt - Install

- Package managers :
  - Pip : Preferred Installer Program



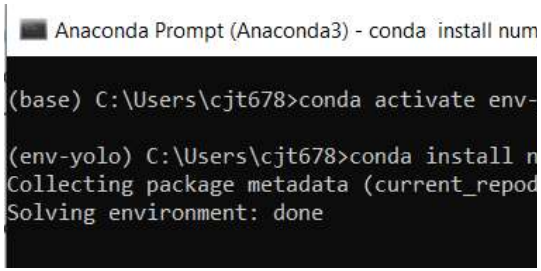
PyPI repository
488k packages
Python only
Example: pip install <package name>

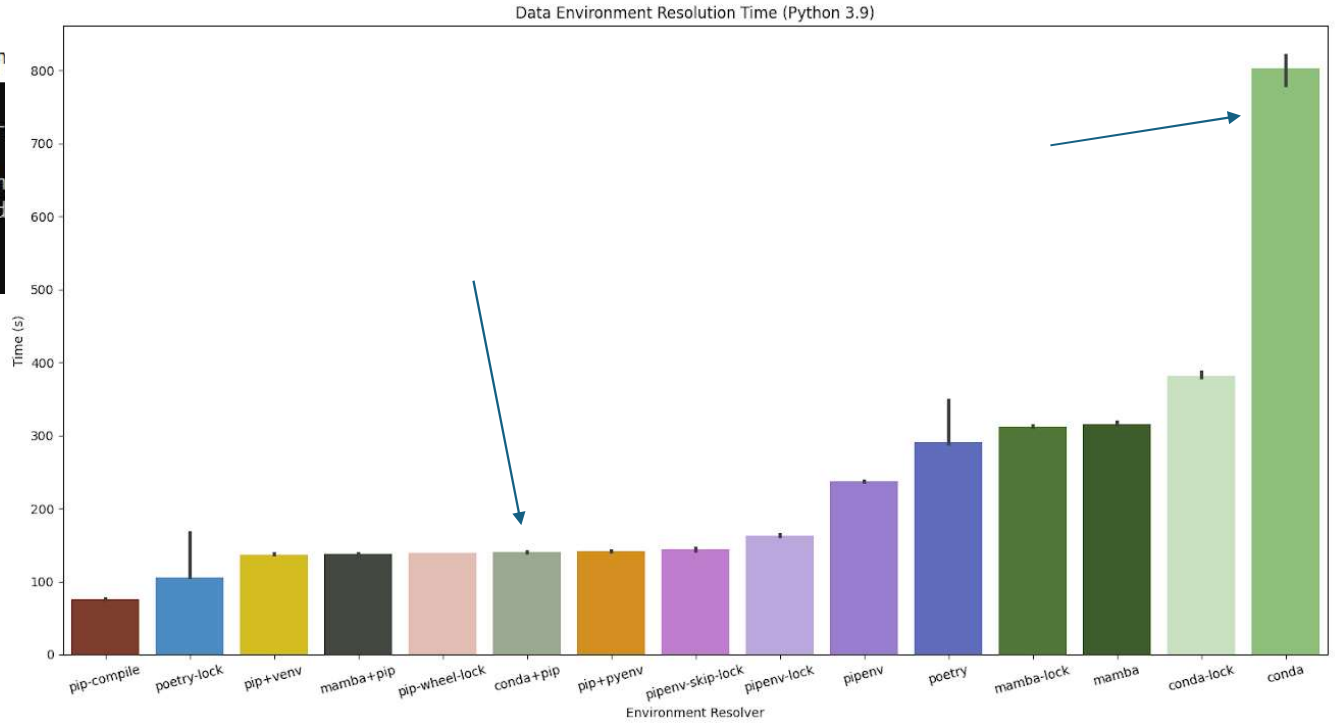
  - Conda



Conda-forge repository
23k packages
Python and other languages
Solver incredibly slow
Example: conda install <package name>

Anaconda Prompt - Install



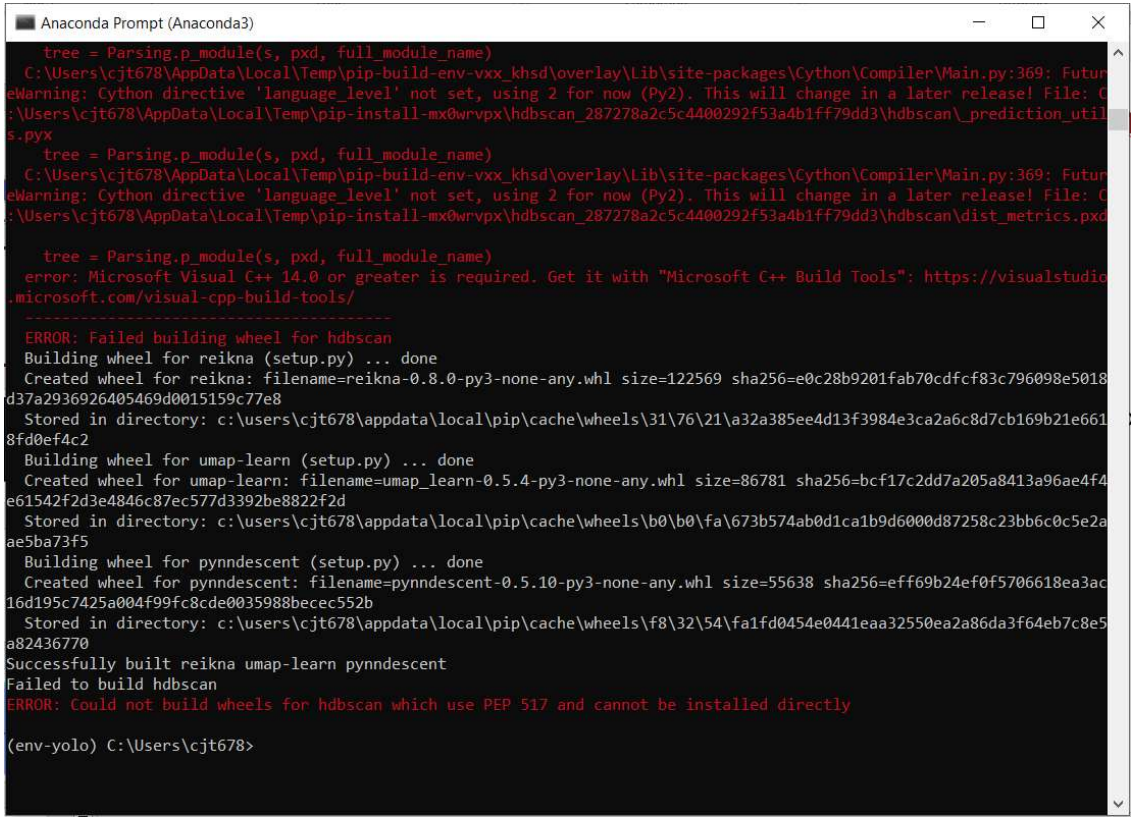


Latest best solver (to my knowledge) – mamba and uv

https://www.recursion.com/news/recursion-developer-experience-evaluating-python-dependency-managers

Anaconda Prompt - Install

Some plugins needs other languages tools
pip install devbio-napari =>



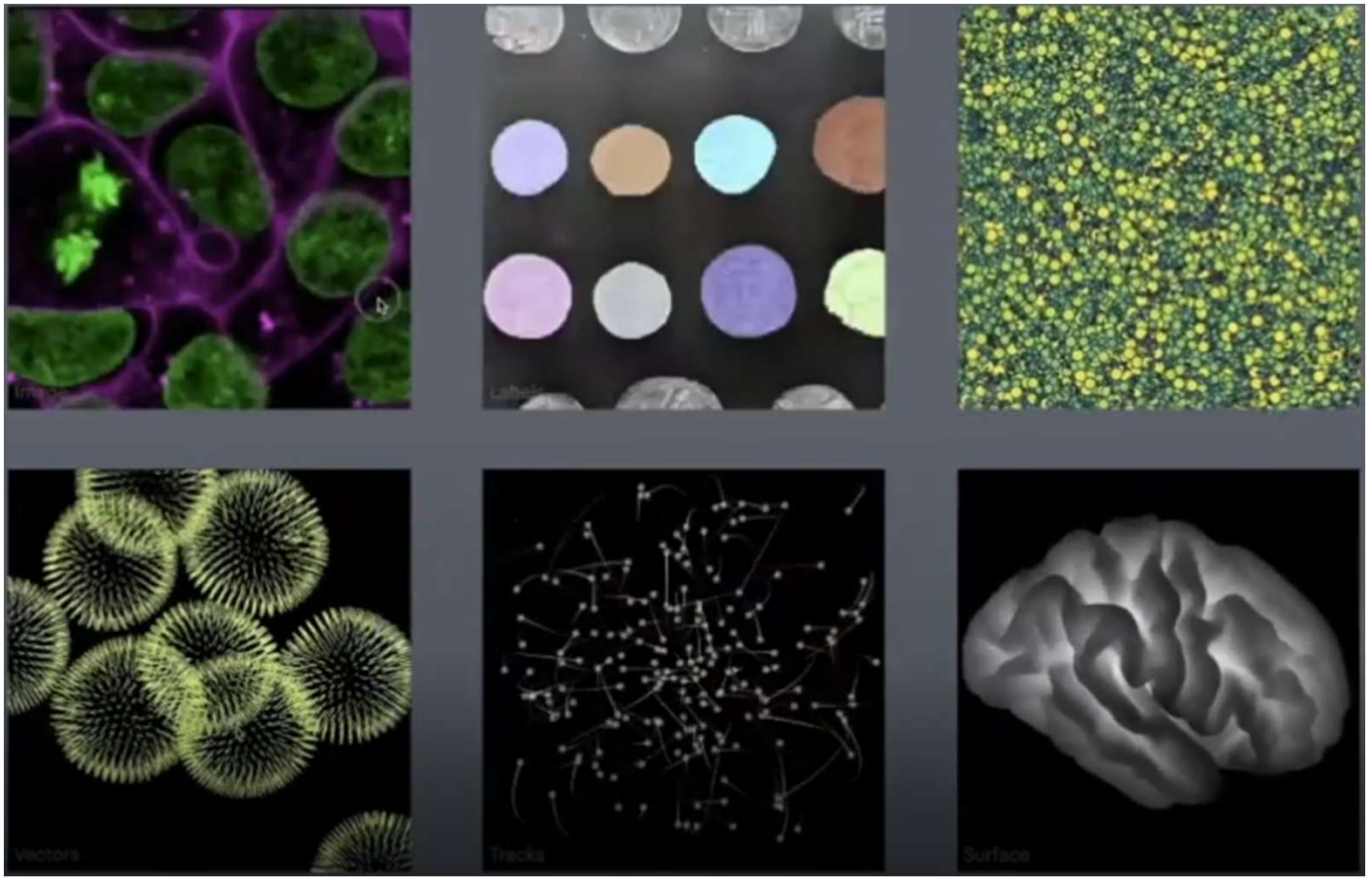
Questions ?

Napari – Basics

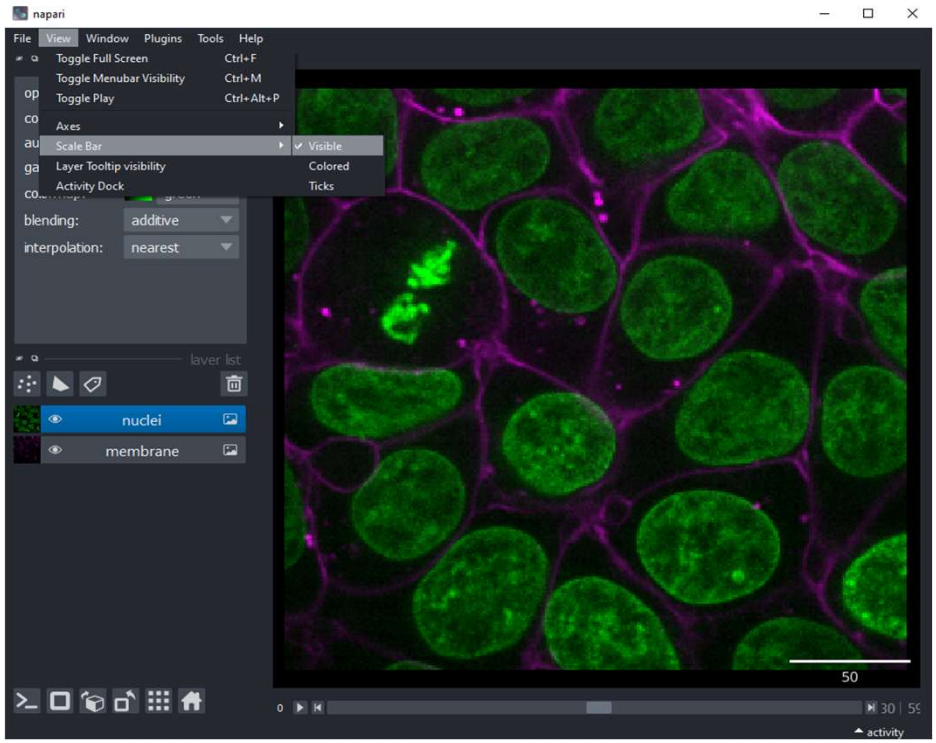
1. Open File > Open Sample > Napari > Cells (3D+2ch)

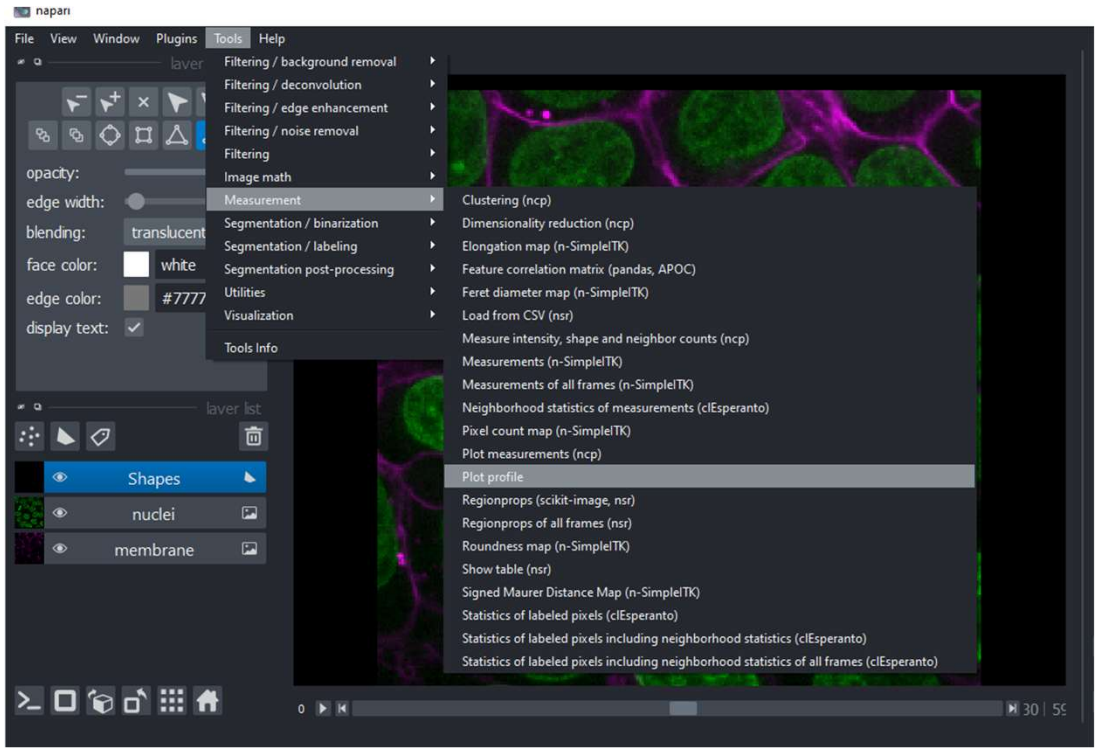


Napari – Layers

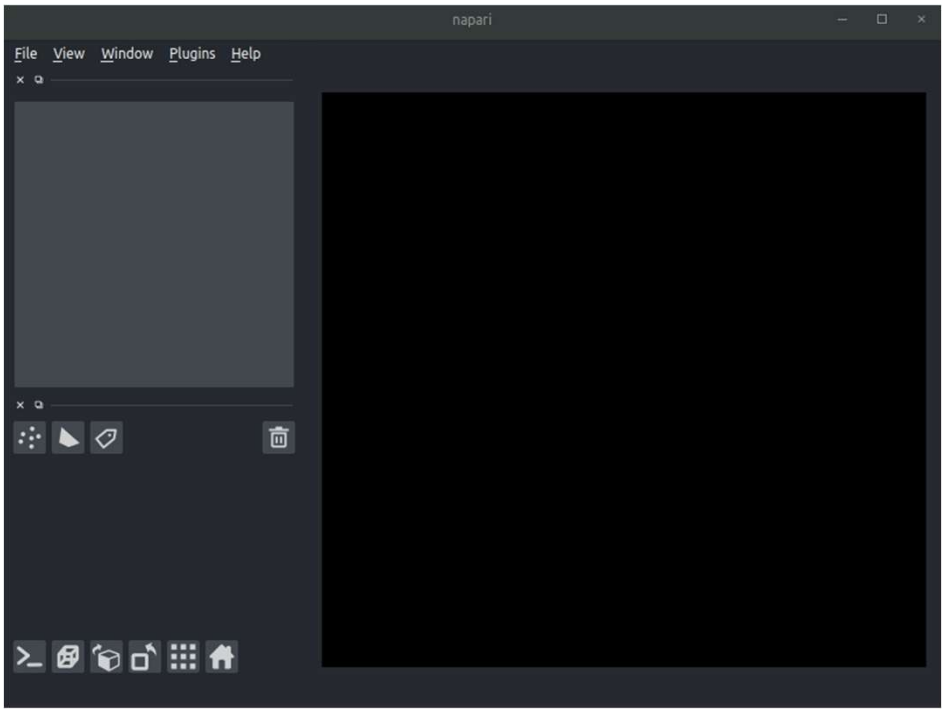


Napari – Scale Bar & Plot Profile

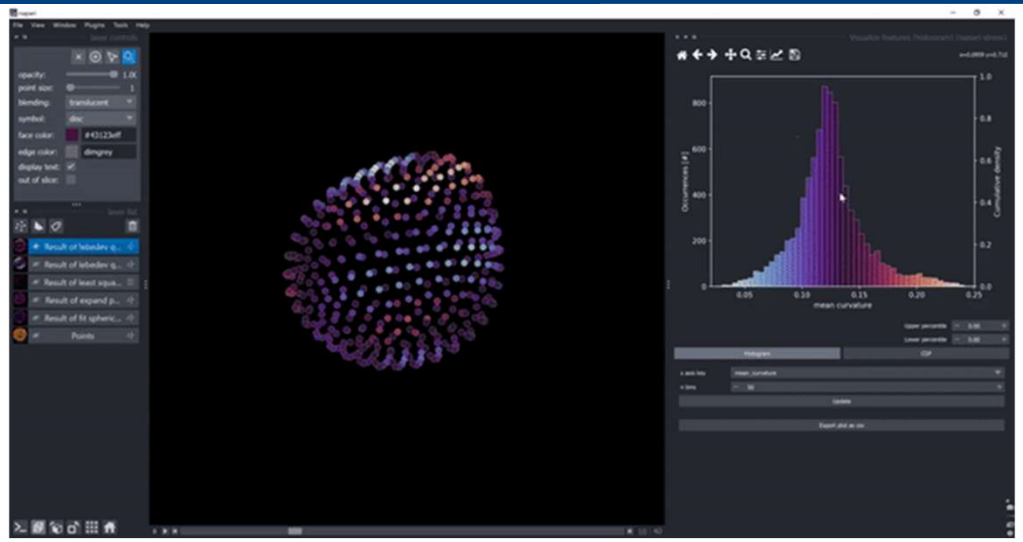




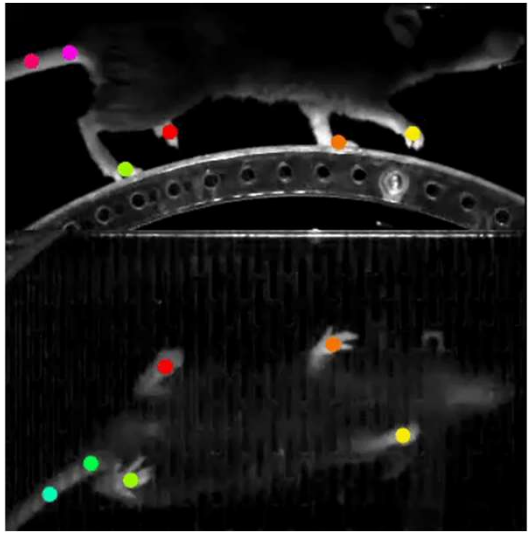
Napari – Examples



Brainreg



Matplotlib

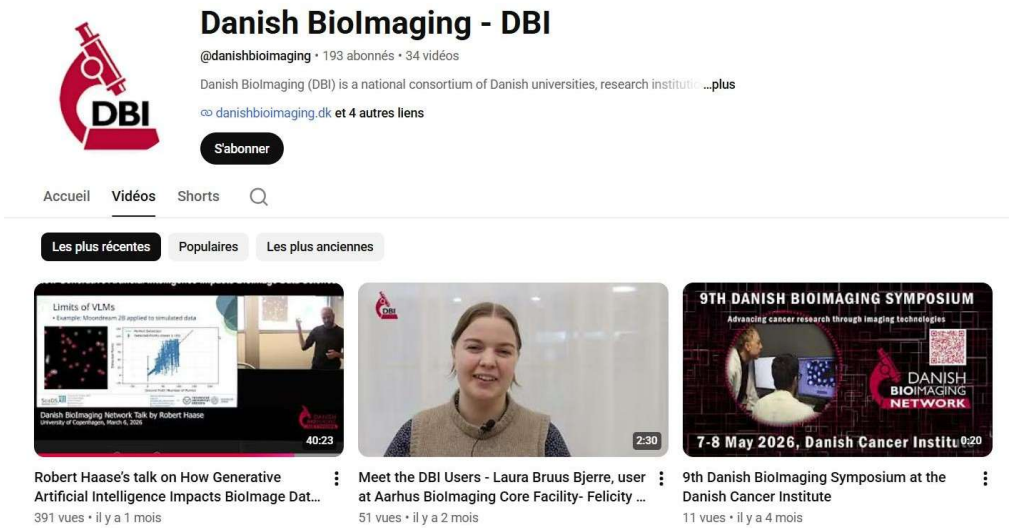


Deeplabcut

Questions ?

Large Language Models





https://www.youtube.com/@danishbioimaging

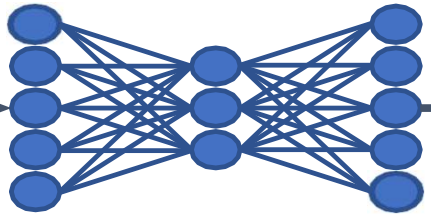


Robert Haase, University of Dresden (Germany)

Large language models

- generate text from text

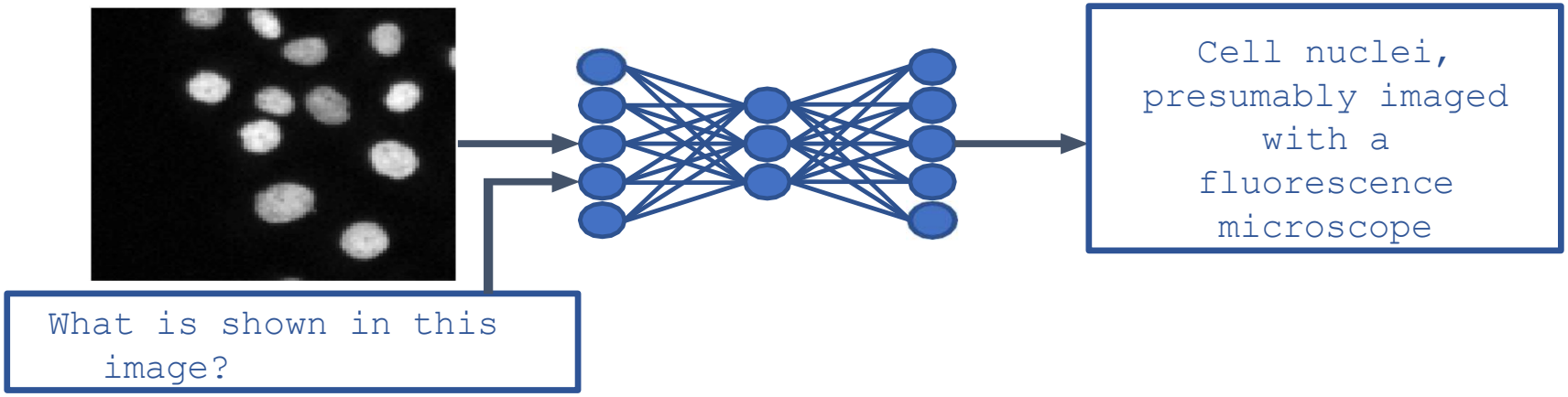
How do fluorescence microscopes work?



A fluorescence microscope is a type of optical microscope that detects light emitted by fluorescent molecules...

Vision language models

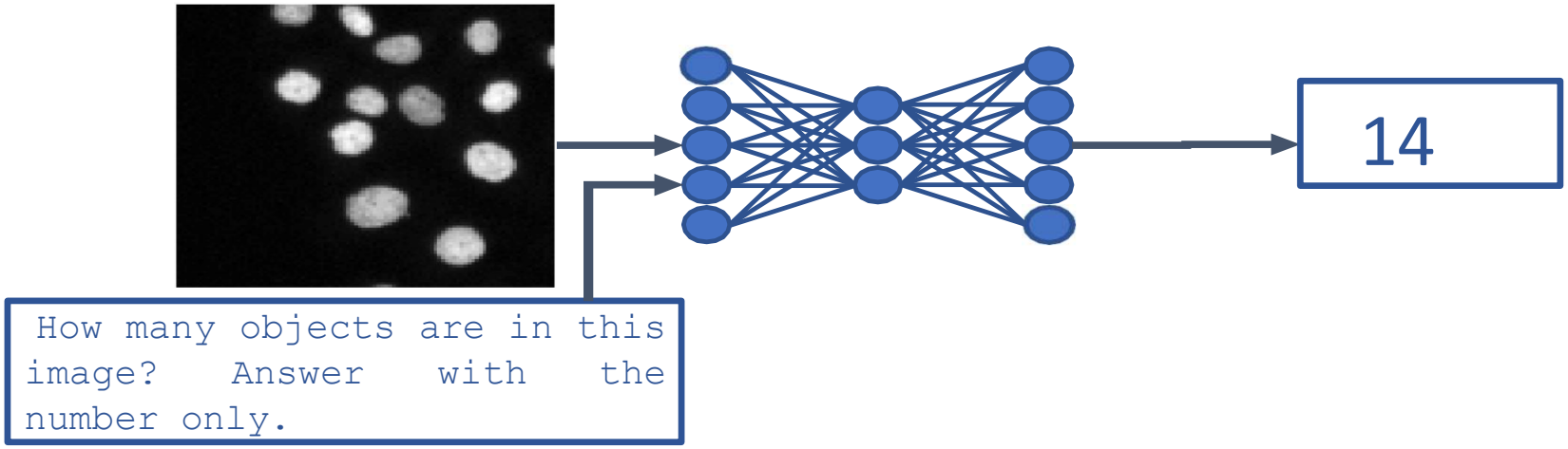
- can describe images



https://doi.org/10.5281/zenodo.18879905

Vision language models

- can count objects

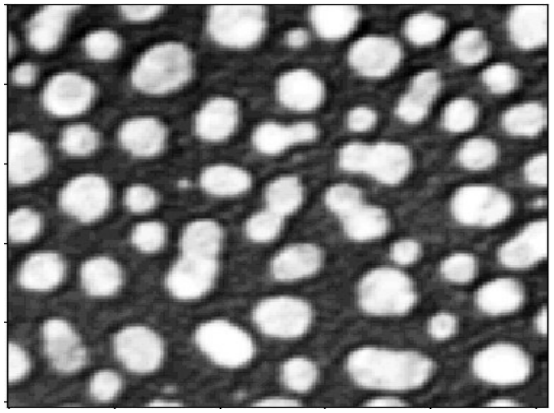


https://doi.org/10.5281/zenodo.18879905

Vision language models for counting objects

- Experiment reproduced June 24th 2025

Lesson learned: We need physical control of models and infrastructure to secure reproducibility!

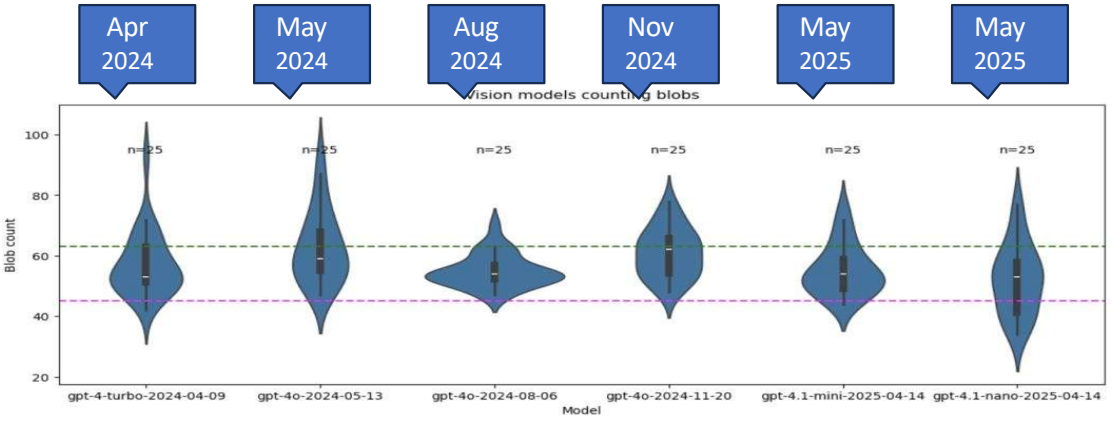


Nov 2023

Model no longer available (Apr. 2025)

Baseline: Otsu-threshold

Baseline: Otsu-threshold, excl. borders

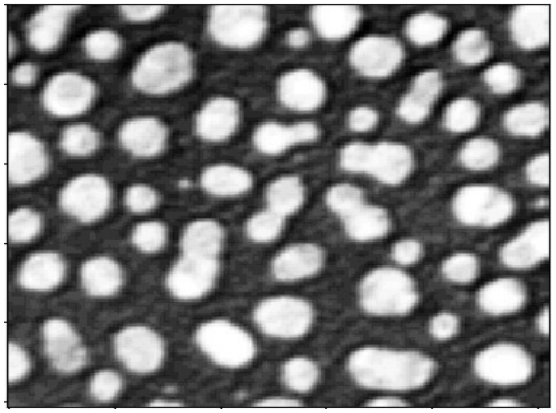


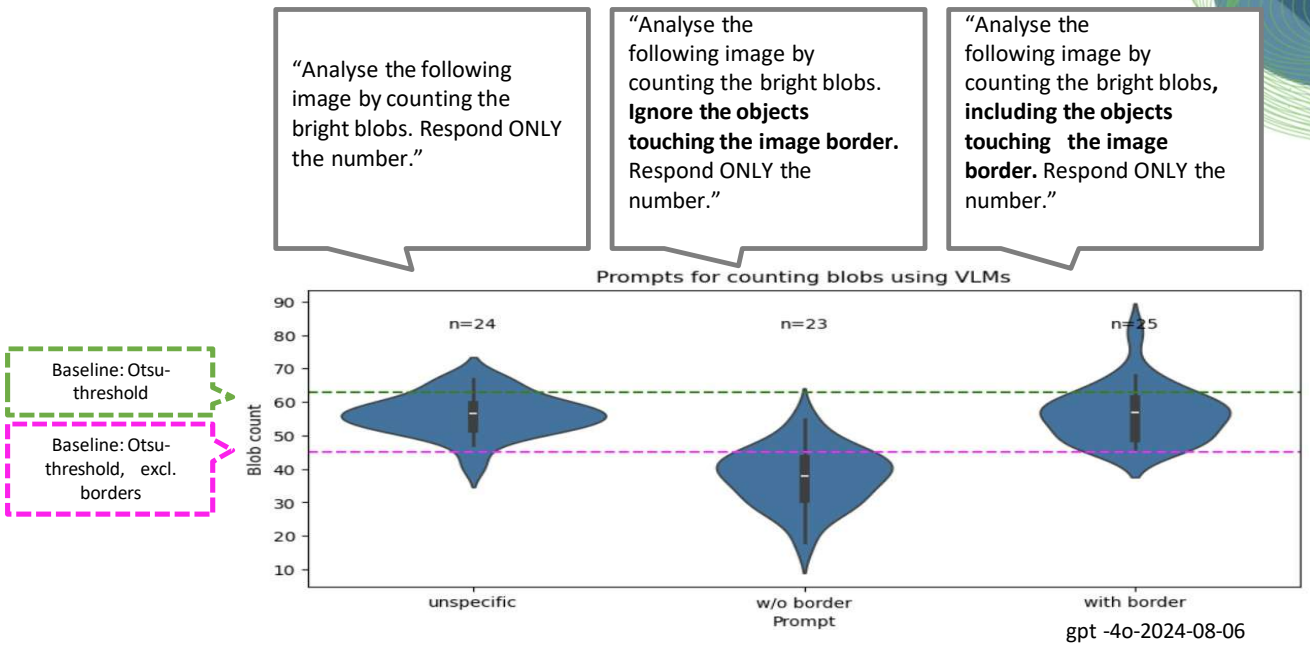
https://scads.github.io/generative-ai-notebooks/80_benchmarking_llms/20_vision_models.html

https://doi.org/10.5281/zenodo.18879905

Vision language models for counting objects

- Prompt-engineering works with VLMs





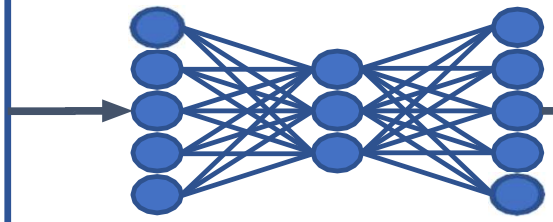
gpt -4o-2024-08-06

https://scads.github.io/generative-ai-notebooks/80_benchmarking_llms/20_vision_models.html

Code generating language models

- Translating from human language to computer language

How can I segment nuclei in a fluorescence microscopy image using Python?



```
from skimage.io import imread, imshow
from skimage.filters import threshold_otsu

# load image
image = imread('your_image.tif')

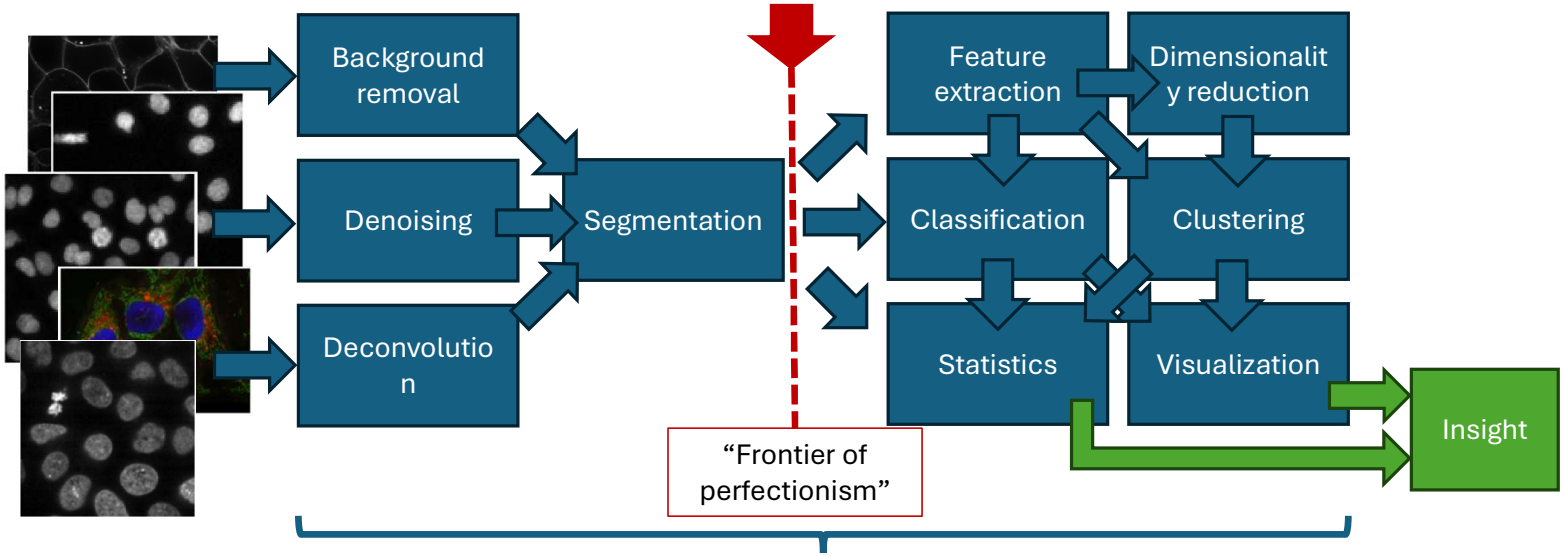
# segment image binary_image = image >
threshold_otsu(image)

# show result imshow(binary_image)
```


https://doi.org/10.5281/zenodo.18879905

Bio-image analysis beyond segmentation

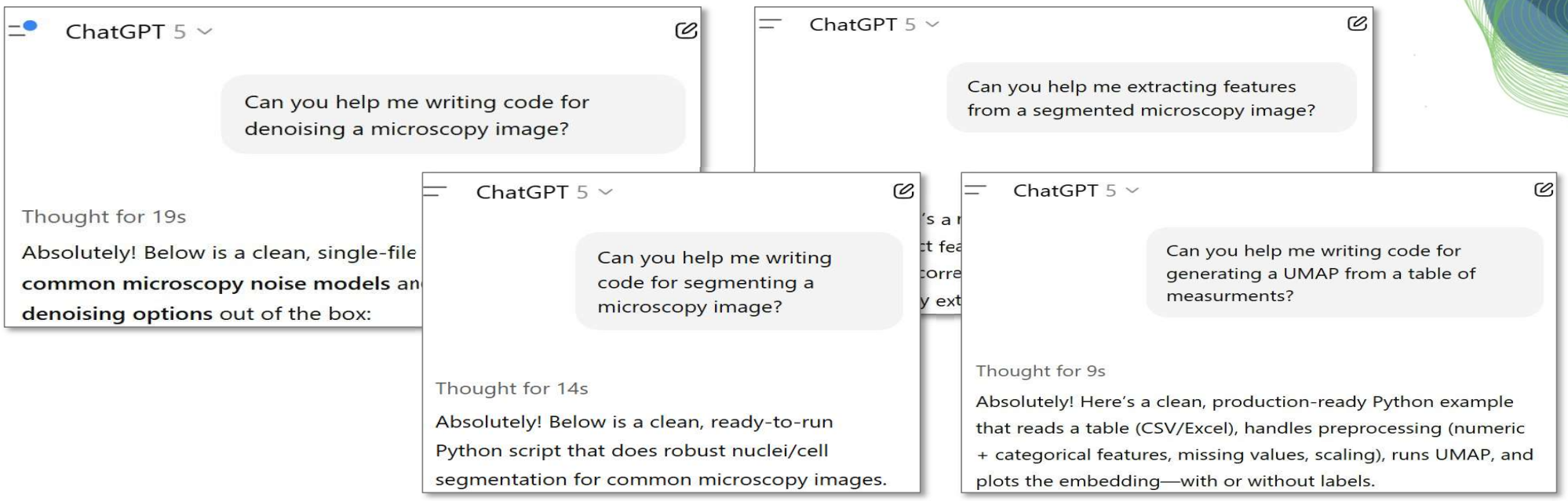
- Crossing this border seems challenging



Wide variety of [computational] skills required to achieve all this

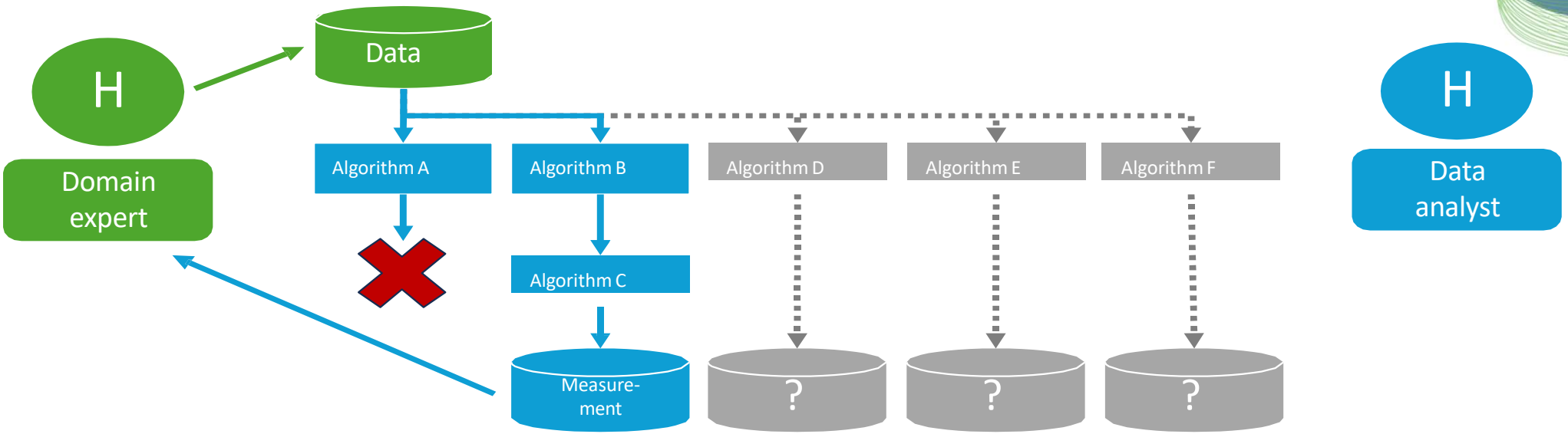
https://doi.org/10.5281/zenodo.18879905

Consulting LLMs



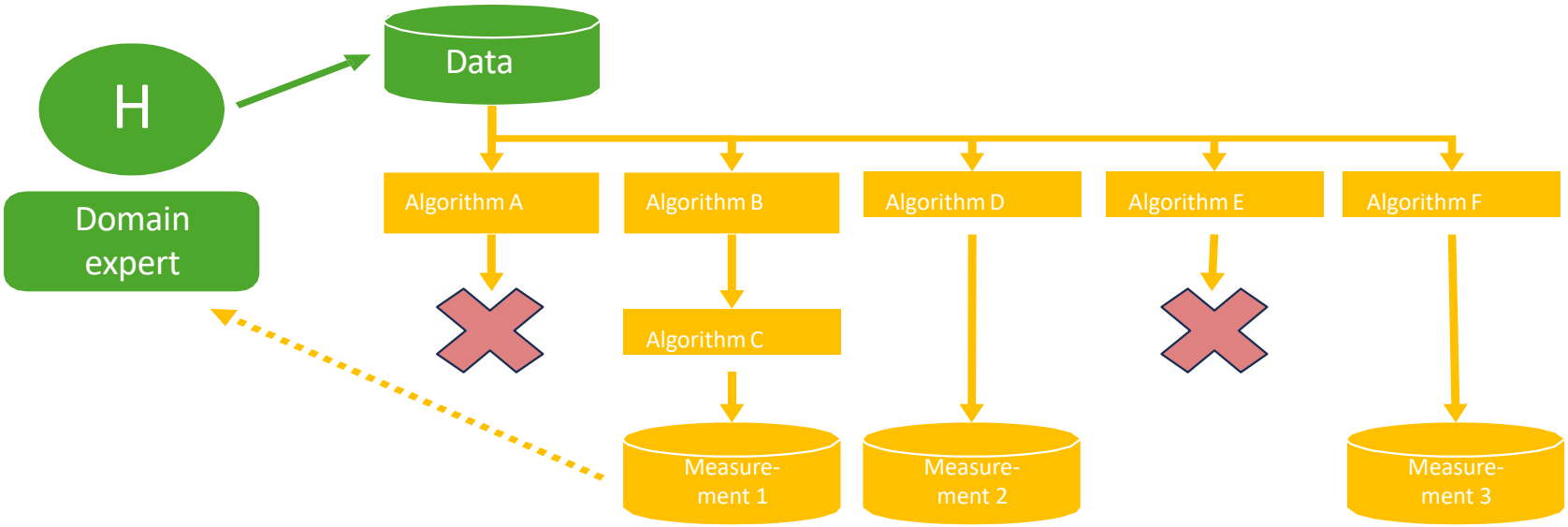
[Image] Data Analysis in a nutshell

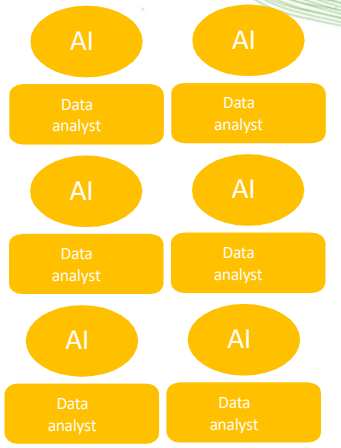
- Data analysts typically go ahead until they find *one* way to analyse data.



AI Agent crowds analysing images in parallel

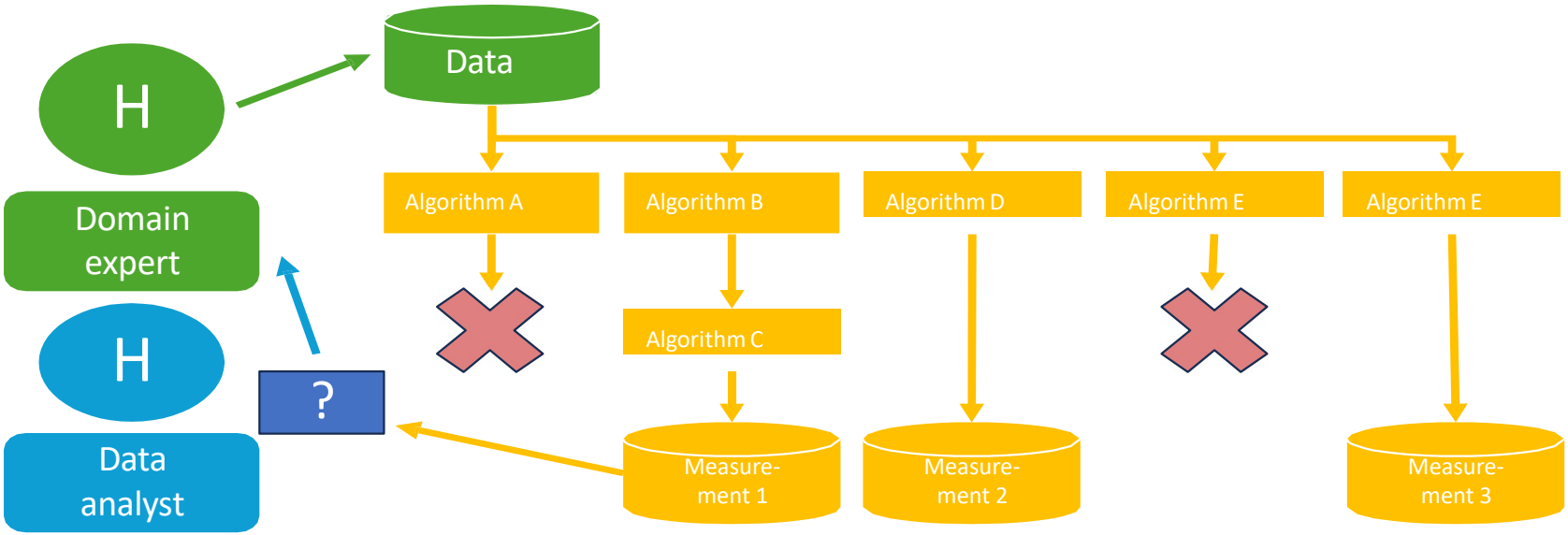
- AI-systems have the "time" to try more analysis methods.

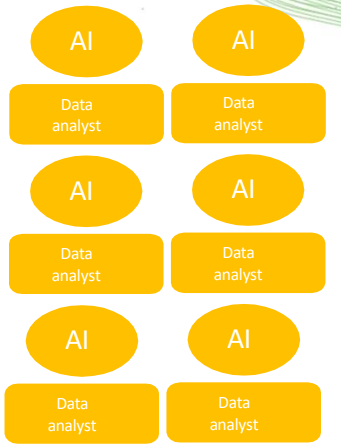




AI Agent crowds analysing images in parallel

- Peer-review of AI-generated content is critical





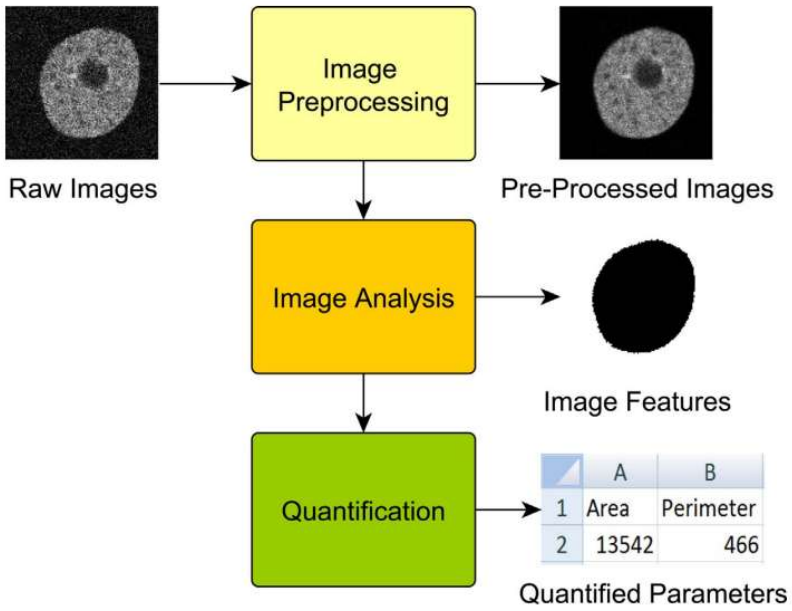
Summary

- Using AI / LLMs for [bioimage] data analysis code generation seems increasingly *unavoidable*
- Maintaining good scientific practice is challenging (esp. peer-review)
  - Compare LLMs / service providers
  - Learn how to use AI *responsibly*
  - Apply good scientific practice (question methods!)

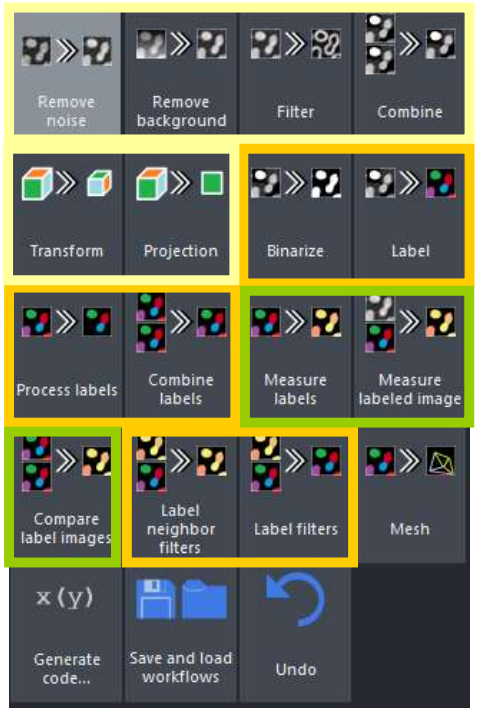
- We may be on the edge towards AI-agents analyzing image data autonomously. https://doi.org/10.5281/zenodo.18879906

Typical Image Analysis workflow :

- Preprocessing – Denoising, normalization
- Segmentation – Thresholding, Machine/Deep Learning
- Measurement



- Restore: denoising, background removal, deconvolution
- Detect/Segment: binarization, labels, object classification
- Measure: shape, intensities, distances



Wollmann & al, Journal of Biotechnology (2017) doi.org/10.1016/j.jbiotec.2017.07.019

https://github.com/haesleinhuepf/devbio-napari

Lost in translation

Fiji / Macro

```
// Gaussian blur filter
run("Gaussian Blur...", "sigma=3");

// Otsu's Threshold
setAutoThreshold("Otsu dark");
setOption("BlackBackground", true);
run("Convert to Mask");

// Connected components labeling
run("Analyze Particles...", "  show=[Count Masks]");
```

MATLAB

```
% Gaussian Blur filter
blurred_image =  imgaussfilt(blobs_image,3)

% Otsu's Threshold
threshold_value = graythresh(blurred_image);
binary_image = blurred_image > threshold_value;

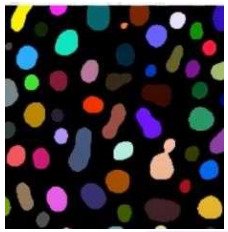
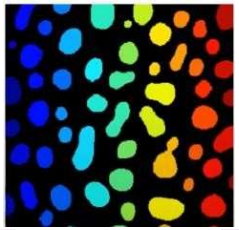
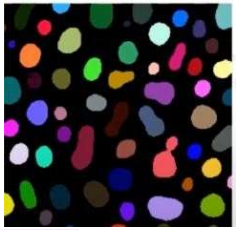
% Connected components labeling
label_image = bwlabel(binary_image);
```

Scikit-Image / Python

```
# Gaussian filter
blurred_image = gaussian(blobs_image, sigma=3)

# Otsu's Thresholding
threshold_value = threshold_otsu(blurred_image)
binary_image = blurred_image > threshold_value

# Connected components labeling
label_image = label(binary_image)
```





Slide courtesy: S. Rigaud